

TSC: Topology-Conditioned Stackelberg Coordination for Multi-Agent Reinforcement Learning in Interactive Driving

Abstract—Safe and efficient autonomous driving in dense traffic is fundamentally a decentralized multi-agent coordination problem, where interactions at conflict points such as merging and weaving must be resolved reliably under partial observability. With only local and incomplete cues, interaction patterns can change rapidly, often causing unstable behaviors such as oscillatory yielding or unsafe commitments. Existing multi-agent reinforcement learning (MARL) approaches either adopt synchronous decision-making, which exacerbates non-stationarity, or depend on centralized sequencing mechanisms that scale poorly as traffic density increases. To address these limitations, we propose Topology-conditioned Stackelberg Coordination (TSC), a learning framework for decentralized interactive driving under communication-free execution, which extracts a time-varying directed priority graph from braid-inspired weaving relations between trajectories, thereby defining local leader–follower dependencies without constructing a global order of play. Conditioned on this graph, TSC endogenously factorizes dense interactions into graph-local Stackelberg subgames and, under centralized training and decentralized execution (CTDE), learns a sequential coordination policy that anticipates leaders via action prediction and trains followers through action-conditioned value learning to approximate local best responses, improving interaction stability and safety in dense traffic. Notably, the learned coordination induces a clear and interpretable yielding-and-passing sequence, which is further demonstrated in real-world proof-of-concept trials under communication-free decentralized execution.

Note to Practitioners: This work is motivated by the deployment of autonomous vehicles and mobile robots in dense, unstructured environments where right-of-way rules are ambiguous, such as crowded intersections and automated warehouses. Conventional solutions often rely on conservative safety rules that may reduce traffic efficiency, or on centralized scheduling and communication infrastructure to negotiate passage priorities. TSC instead enables each agent to infer local leader–follower relations from onboard observations and execute a decentralized policy without explicit communication or a global priority order. Experimental results show that TSC substantially reduces collisions while maintaining traffic efficiency and control smoothness across multiple dense interaction scenarios. Its policy inference latency remains within the control interval used in our experiments, supporting real-time decentralized execution. The method is particularly relevant to bandwidth-constrained autonomous-vehicle and multi-robot systems. Nevertheless, practical deployment still requires accounting for sensing, state-estimation, and actuation delays, as well as robustness to perception uncertainty and heterogeneous traffic participants.

Index Terms—Multi-agent systems, Reinforcement learning, Decentralized control, Stackelberg game

I. INTRODUCTION

AUTONOMOUS vehicles (AVs) must coordinate safely and efficiently in dense traffic, which is a key capability

for next-generation intelligent transportation systems [1]–[7]. In contrast to single-vehicle planning, multi-vehicle coordination must satisfy safety and kinematic constraints while forming consistent right-of-way patterns in shared spaces such as intersections, merges, and on-ramps [8]–[16]. Recently, MARL has emerged as a practical paradigm for modeling such interactions, enabling distributed agents to learn coordination strategies through trial and error across diverse transportation tasks [8], [17]–[20]. This paradigm is particularly attractive under decentralized execution, where accurate system dynamics are difficult to obtain, communication between vehicles and infrastructure is constrained, and relying on long-horizon centralized global information is often impractical [21]–[25]. Nevertheless, learning stable right-of-way patterns from local observations remains challenging in dense conflict regions, where multiple vehicles compete for the same spatiotemporal gaps and must implicitly negotiate a passage order without communication. Reciprocal reaction loops and concurrent policy updates then render the effective game highly non-stationary, often manifesting as oscillatory yielding, deadlocks, or unsafe commitments [26], [27].

From a game-theoretic perspective, decentralized coordination can be cast as a partially observed dynamic non-cooperative game, where agents choose control actions to optimize safety–efficiency objectives under coupled collision-avoidance constraints, and stable behavior corresponds to an equilibrium that implicitly defines a consistent right-of-way ordering [12], [14], [28]. According to decision timing, existing MARL-based coordination methods can be broadly grouped into two classes. The first class formulates coordination as a simultaneous-move (joint-action) Markov game, where all agents select actions synchronously at each time step [1], [8]–[10]. Since agents update their policies concurrently during learning, the induced environment becomes non-stationary from any individual agent’s viewpoint, which can destabilize training and hinder convergence [17]–[19], [23], [25]. The second class introduces sequential decision structures or priority mechanisms to impose an explicit or implicit order of play, approximating Stackelberg-style leader–follower interactions in which some agents commit earlier and others respond, thereby encouraging more consistent and stable coordination patterns [12], [13], [26]–[30]. However, many such approaches depend on constructing a global order of play or unrolling a fixed multi-agent decision sequence, causing representational and computational costs to grow quickly with the number of vehicles and limiting scalability in large, heterogeneous dense traffic settings.

Topological structures have been widely used to describe multi-agent interaction patterns that arise from trajectory weaving, merging, and overtaking [31]–[33]. In particular,

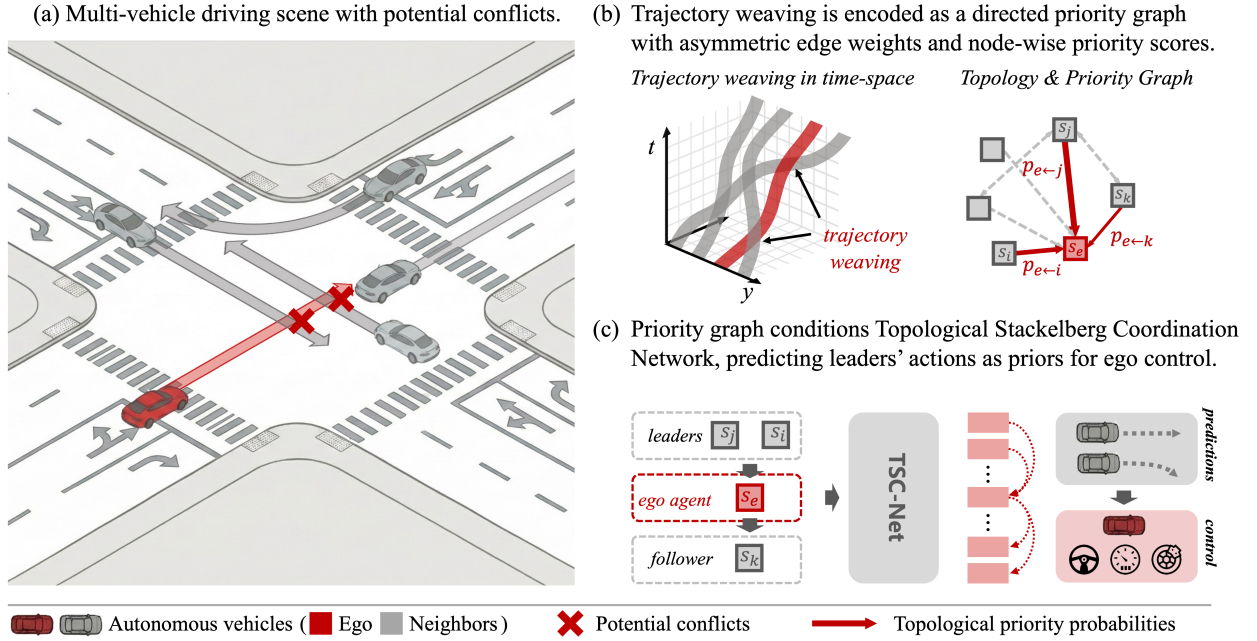


Fig. 1: Topology-conditioned Stackelberg coordination from trajectory weaving in multi-vehicle traffic.

braid-theoretic formulations abstract crossings and precedence relations into topological braids or directed graphs, where the crossing order of intertwined trajectories induces a directed precedence structure that supports compact reasoning about interaction complexity and priority constraints in multi-robot coordination [31], [32]. In autonomous driving, related braid-inspired abstractions have been adopted to organize multi-vehicle interaction prediction and planning, for example by grouping joint futures into topological equivalence classes based on relative crossing order, or by leveraging lane connectivity and homotopy cues to obtain more consistent and rule-compliant behaviors [33]–[36]. These studies demonstrate that trajectory topology provides a compact and interpretable description of interaction structure. More specifically, Mavrogiannis et al. [31], [32] primarily use braid-based abstractions to represent and analyze crossing order, interaction complexity, and priority relations, whereas Liu et al. [33] employ behavioral topology to organize integrated trajectory prediction and planning. These works establish topology as an informative representation of multi-agent interactions, but do not embed it into decentralized policy learning to induce local orders of play.

Rule-based intersection management and communication-enhanced driving systems introduce ordering or information-sharing mechanisms, such as reservation-based priority assignment and V2X-assisted perception [37]–[40]. V2X-based methods such as V2X-Lead [39] mitigate partial observability by augmenting onboard perception with communicated information. In contrast, TSC infers local priority relations from ego-centric observations without requiring communication; when V2X is available, its richer observations can further support topology-conditioned priority inference. Despite these advances, topology is typically introduced as an external representation [31], [33], rather than being embedded into the underlying multi-agent decision structure to induce fea-

sible orders of play and directly guide policy learning under communication-constrained decentralized execution [21]–[24]. Thus, how to transform trajectory-level topological relations into decision-relevant leader–follower dependencies for communication-free decentralized policy learning remains insufficiently explored.

To address these limitations, we propose Topology-conditioned Stackelberg Coordination (TSC), a multi-agent learning framework for safe and efficient cooperative interactive driving in dense traffic under communication-free execution. In shared conflict regions, interactions are governed by an evolving local precedence structure, whereas enforcing a global order of play is brittle and scales poorly with increasing traffic density [12], [14], [26]–[30]. Instead, TSC infers a time-varying directed priority graph from braid-inspired trajectory weaving cues, which reflect prospective crossing order and the strength of interaction coupling from ego-centric observations. This graph provides a compact interaction scaffold. Using this graph, each agent selects a limited number of interaction-critical neighbors to form a sparse decision state, and the resulting directed relations define local leader–follower dependencies that support graph-local Stackelberg reasoning rather than a centralized decision sequence.

TSC differs from prior topology-based methods in how topology is integrated into decision making. While Mavrogiannis et al. [31], [32] use braid topology to represent and analyze interaction structures, and Liu et al. [33] employ behavioral topology to organize prediction and planning, TSC embeds an online-inferred priority graph into MARL to sparsify interactions, define local leader–follower relations, and condition critic-side value estimation. This design provides a scalable coordination structure without requiring a global order of play or inter-agent communication. By coupling priority-guided sparsification with local sequential reasoning, TSC mitigates reciprocal reaction loops in dense interactions and concentrates

policy learning on safety-critical conflicts. To implement TSC, we propose a topology-conditioned priority inference and policy learning network (TSC-Net) trained under centralized training and decentralized execution. TSC-Net jointly learns priority graph inference and topology-conditioned policy, with consistency-driven supervision that links edge-wise priority probabilities with score-induced precedence so that inferred priorities remain stable and decision-relevant. At execution time, each agent acts only from compact priority-filtered local observations, enabling communication-free yet coherent leader–follower behavior.

Contributions. We summarize our contributions as follows:

- We propose TSC, a Topology-conditioned Stackelberg Coordination framework for dense multi-vehicle interaction under communication-free decentralized execution, which infers a time-varying directed priority graph from braid-inspired weaving cues to induce graph-local leader–follower dependencies without constructing a global order of play.
- We develop TSC-Net, a CTDE actor-critic instantiation of TSC that jointly learns edge-wise priority probabilities and node-wise priority scores with a consistency regularizer, and conditions value learning on predicted leader actions to enable consistent local sequential reasoning without explicit communication.
- We evaluate TSC in four dense traffic scenarios and in a three-robot real-world proof-of-concept experiment, showing consistent collision reduction and improved interaction stability, with ablations that validate the impact of priority inference, topology-guided sparsification, and local Stackelberg reasoning.

II. PROBLEM FORMULATION

A. Task and Objective

We consider an interactive driving scenario with $N \geq 2$ connected and automated vehicles evolving in discrete time [9], [41], [42]. Each vehicle is controlled by an agent and we use the same index i for both. Let $\mathcal{I} = \{1, \dots, N\}$ and $\mathcal{T} = \{1, \dots, T\}$ denote the agent set and episode time indices.

We model the problem as a partially observable stochastic game with tuple:

$$\langle N, \mathcal{S}, \{\mathcal{A}_i\}_{i=1}^N, \{\mathcal{O}_i\}_{i=1}^N, \{\Omega_i\}_{i=1}^N, \mathcal{P}, \{r_i\}_{i=1}^N, T, \gamma \rangle, \quad (1)$$

where $\mathcal{S} \subseteq \mathbb{R}^{d_s}$ is the global state space and $\mathcal{A}_i = \mathcal{A} \subseteq \mathbb{R}^{d_a}$. At time step t , the system is at a global state $s^t \in \mathcal{S}$. Each agent i chooses an action $a_i^t \in \mathcal{A}_i$ and receives a local observation $o_i^t \in \mathcal{O}_i$ given by $o_i^t = \Omega_i(s^t)$, where Ω_i maps the global state to the observable information of agent i . The stochastic transition function is denoted by $\mathcal{P} : \mathcal{S} \times \mathcal{A}_1 \times \dots \times \mathcal{A}_N \rightarrow \Delta(\mathcal{S})$, and the next state is sampled as $s^{t+1} \sim \mathcal{P}(s^t, a_1^t, \dots, a_N^t)$. Agent i receives a scalar reward $r_i^t = r_i(s^t, a_1^t, \dots, a_N^t)$, and $\gamma \in (0, 1)$ is the discount factor.

Local observation. At time t , the set of agents observable to agent i is denoted by $\mathcal{N}_i^t \subseteq \{1, \dots, N\} \setminus \{i\}$. For implementation, $\mathcal{N}_i^t$ is represented as a fixed-size candidate set, padding when fewer are available. The local observation

is represented as $o_i^t = \{\tilde{o}_i^t, \{o_{i,j}^t\}_{j \in \mathcal{N}_i^t}\}$, where $\tilde{o}_i^t$ collects ego-related features of agent i and $o_{i,j}^t$ encodes features of neighbor j as seen from agent i . This can be obtained from on-board sensing without inter-agent communication.

We instantiate the state, observation, and action spaces using the VMAS [43] simulation environment following the setup of SigmaRL [9], [23]. Each agent is modeled as a nonlinear kinematic bicycle model with continuous speed and steering-angle control. The self-observation $\tilde{o}_i^t$ and neighbor observation $o_{i,j}^t$ are constructed from structured distance-based features rather than raw sensor data, and the reward function r_i combines safety, efficiency, path-following, and smoothness terms. The full specification of the state, observation, action spaces, and reward function is given in Appendix B.1.

Policy and objective. Each agent follows a decentralized policy $\pi_\theta : o_i \rightarrow \Delta(\mathcal{A}_i)$ that maps its local observation to an action distribution. Under CTDE with parameter sharing [44], [45], we optimize shared parameters θ to maximize the average discounted return

$$\theta^* = \arg \max_{\theta} \mathbb{E}_{\pi_\theta} \left[\sum_{t=1}^T \gamma^t \frac{1}{N} \sum_{i=1}^N r_i(s^t, a_1^t, \dots, a_N^t) \right]. \quad (2)$$

This setting is challenging due to non-stationarity induced by mutual reactions and the need for consistent implicit ordering in dense conflict regions.

B. Topological Interaction Graph

In dense interactions, purely distance-based neighborhoods may be insufficient to capture behaviorally coupled relations such as merging or crossing. We therefore introduce a directed topological interaction graph that models local precedence and influence as directed dependencies among agents and supports priority-aware coordination under decentralized execution. At time t , the scene is represented by a directed graph

$$G_t = (V_t, E_t, W_t), \quad (3)$$

where each node $v_i^t \in V_t$ corresponds to agent i with node feature $x_i^t \in \mathbb{R}^{d_x}$. For $j \neq i$, an edge $(j, i) \in E_t$ indicates that agent j can constrain the feasible maneuvers of agent i and carries a nonnegative weight $w_{i \leftarrow j}^t \in \mathbb{R}_{\geq 0}$ that is not necessarily symmetric and will be instantiated as a directed priority strength in our construction. To respect partial observability, edge features $z_{i \leftarrow j}^t \in \mathbb{R}^{d_z}$ are computed from quantities available to agent i via on-board sensing. Under decentralized execution, agent i constructs an ego-centric local view G_i^t induced by $i \cup \mathcal{N}_i^t$ and performs subsequent inference and decision making from G_i^t using only this local view without global scene access or communication [33].

III. METHOD

Fig. 1 summarizes the proposed framework and its instantiation in TSC-Net. Sec. III-A formalizes the trajectory weaving-based priority formulation and constructs a directed priority graph with asymmetric edge weights that encodes local precedence among potentially conflicting vehicles. Sec. III-B then presents TSC-Net, which infers an ego-centric local

priority graph and performs topology-conditioned Stackelberg coordination using predicted leader actions to support communication-free execution. Sec. III-C describes the learning objective and the CTDE training procedure.

A. Topological Priority Formulation

1) *Agent-centric future-trajectory representation*: Let $\mathcal{Y}^t = \{\mathbf{Y}_i^t\}_{i=1}^N$ denote the multi-agent future trajectories over a finite horizon H , where $\mathbf{Y}_i^t = \{\mathbf{X}_i^{t+h}\}_{h=0}^H \in \mathbb{R}^{(H+1) \times 2}$ and $\mathbf{X}_i^{t+h} = [p_{x,i}^{t+h}, p_{y,i}^{t+h}]^\top \in \mathbb{R}^2$ collects 2D position. These future rollouts are used only for training supervision and are unavailable at decentralized execution. We are interested in how these trajectories intertwine in the neighborhood of each agent and in representing such interactions in a structured form that can be exploited by a graph-based decision model.

For each ego agent i , we introduce an agent-centric lateral-longitudinal coordinate frame. Let $\theta_i^t \in \mathbb{R}$ denote its heading over the horizon and $R_i^t \in \mathbb{R}^{2 \times 2}$ the rotation matrix that aligns the x -axis of this frame with θ_i^t . The future position of any agent j is mapped into this local frame over a horizon H as:

$$\tilde{\mathbf{X}}_j^{(i)}(t+h) = R_i(\mathbf{X}_j^{t+h} - \mathbf{X}_i^t) \in \mathbb{R}^2, \quad h = 0, \dots, H \quad (4)$$

We denote its lateral coordinate by $y_k^{(n)}(t+h)$, and $\{y_j^{(i)}(t+h)\}_{h=0}^H \in \mathbb{R}^{H+1}$ describes, from the viewpoint of agent i , the sideways motion of j relative to i .

2) *Braid-inspired weaving distance*: Consider two distinct agents (i, j) at time t . In the ego-centric lateral-longitudinal frame of agent i , define the lateral gap as:

$$\Delta_{i \leftarrow j}^{(i)}(h) = y_i^{(i)}(t+h) - y_j^{(i)}(t+h), \quad h = 0, \dots, H. \quad (5)$$

Intuitively, strong weaving occurs when the two future lateral profiles approach each other and tend toward a braid crossing, i.e., $\Delta_{i \leftarrow j}^{(i)}(h)$ becomes small in magnitude and exhibits a sign change over consecutive steps. Motivated by braid-theoretic reasoning of multi-agent trajectory topology [46], [47], we use the agent-centric lateral-time projection to describe the relative lateral ordering of two future trajectories. Rather than constructing a full braid word or detecting exact braid crossings, we use variations in the signed lateral gap as a soft geometric cue for potential weaving interactions. This soft weaving cue is then used to define directed precedence between the pair and to supervise the corresponding edge in the priority graph.

To connect this soft weaving cue with a continuous priority score, we define the soft crossing evidence

$$\eta_{i \leftarrow j}^{(i)}(h) = \max(0, -\Delta_{i \leftarrow j}^{(i)}(h)\Delta_{i \leftarrow j}^{(i)}(h+1)), \quad (6)$$

This term becomes positive when the signed lateral gap changes sign between two consecutive steps and remains zero for same-side or near-parallel motion. It therefore provides a soft geometric indicator of possible weaving, rather than a discrete braid generator, a formal braid invariant, or an exact crossing detector.

To quantify this effect, for $h = 0, \dots, H-1$ we define a per-step near-crossing score:

$$\delta_{i \leftarrow j}^{(i)}(h) = \frac{\min(|\Delta_{i \leftarrow j}^{(i)}(h)|, |\Delta_{i \leftarrow j}^{(i)}(h+1)|)}{\varepsilon + \eta_{i \leftarrow j}^{(i)}(h)} \in \mathbb{R}_+, \quad (7)$$

where $\varepsilon > 0$ is a stabilizing constant. The numerator measures the closest lateral separation over two consecutive steps, while the denominator incorporates the soft crossing evidence and yields smaller values when a signed lateral-order exchange occurs (since $-\Delta(h)\Delta(h+1) > 0$). For brevity, we write $\Delta(h)$ for $\Delta_{i \leftarrow j}^{(i)}(h)$ when no ambiguity arises. If the two lateral profiles genuinely cross, then $\Delta(h)\Delta(h+1) < 0$ and $\eta_{i \leftarrow j}^{(i)}(h) > 0$, which reduces $\delta_{i \leftarrow j}^{(i)}(h)$ and indicates stronger weaving. If the trajectories are nearly parallel and remain on the same side, then $\Delta(h)\Delta(h+1) \geq 0$ and $\eta_{i \leftarrow j}^{(i)}(h) = 0$; the score is then determined mainly by the closest lateral separation scaled by ε . For close-but-non-crossing pairs, the small numerator still captures local proximity, but the absence of a sign change means that no additional crossing-order evidence is introduced. The constant ε prevents division by zero and controls the scale of proximity-only cases where no clear crossing evidence is available. We keep ε fixed across all scenarios and examine its sensitivity in the ablation study.

We then aggregate over the horizon by taking the most critical near-crossing or proximity event, defining the directed weaving distance:

$$d_{i \leftarrow j}^t = \min_{h=0, \dots, H-1} \delta_{i \leftarrow j}^{(i)}(h). \quad (8)$$

Smaller $d_{i \leftarrow j}^t$ indicates stronger topological coupling between the future motions of i and j as viewed from agent i 's frame, whereas larger values indicate weak or negligible weaving. Thus, $d_{i \leftarrow j}^t$ is a braid-inspired continuous weaving score. It reflects signed-gap variations in the lateral-time projection while further incorporating geometric proximity and soft crossing evidence. It is not intended to be a full braid word, a formal braid invariant, or a calibrated crossing probability. Repeating the same construction with roles swapped (i.e., in the local frame of agent j) yields $d_{j \leftarrow i}^t$.

3) *Soft pairwise topological priority*: Given the two directed distances, we assign asymmetric pairwise priority probabilities via a two-class softmax:

$$p_{i \leftarrow j}^t = \frac{\exp(-d_{i \leftarrow j}^t/\tau)}{\exp(-d_{i \leftarrow j}^t/\tau) + \exp(-d_{j \leftarrow i}^t/\tau)}, \quad p_{j \leftarrow i}^t = 1 - p_{i \leftarrow j}^t, \quad (9)$$

where $\tau > 0$ controls the softness. Equivalently, this probability can be written as

$$p_{i \leftarrow j}^t = \sigma\left(\frac{d_{j \leftarrow i}^t - d_{i \leftarrow j}^t}{\tau}\right), \quad (10)$$

where $\sigma(\cdot)$ is the logistic sigmoid. This formulation has three useful properties for constructing pairwise priority edges. First, the reciprocity $p_{i \leftarrow j}^t + p_{j \leftarrow i}^t = 1$ makes the priority a relative relation between two agents and avoids contradictory dominance assignments on the same pair. Second, the order consistency $p_{i \leftarrow j}^t > 1/2 \Leftrightarrow d_{i \leftarrow j}^t < d_{j \leftarrow i}^t$ ensures that the probability preserves the directed ordering induced by the two weaving distances, making the priority assignment directly traceable to the geometric weaving scores. Third, the monotonicity

$$\frac{\partial \delta_{i \leftarrow j}^{(i)}(h)}{\partial \eta_{i \leftarrow j}^{(i)}(h)} \leq 0, \quad \frac{\partial p_{i \leftarrow j}^t}{\partial d_{i \leftarrow j}^t} < 0, \quad \frac{\partial p_{j \leftarrow i}^t}{\partial d_{j \leftarrow i}^t} > 0 \quad (11)$$

implies that stronger soft weaving evidence reduces the corresponding directed distance and increases the associated priority probability. These properties make $p_{i \leftarrow j}^t$ a smooth, direction-consistent, and interpretable priority score derived from the weaving-distance construction.

With the arrow convention $i \leftarrow j$ (meaning influence from j to i), a larger $p_{i \leftarrow j}^t$ indicates that j is more dominant relative to i within their intertwined region. Collecting agents as vertices V_t , the pairwise probabilities $\{p_{i \leftarrow j}^t\}$ define the edge weights of the topological interaction graph $G_t = (V_t, E_t, W_t)$ introduced in Sec. II-B, by setting $w_{i \leftarrow j}^t = p_{i \leftarrow j}^t$ for each interaction relevant ordered pair $(j, i) \in E_t$. The general asymmetry $p_{i \leftarrow j}^t \neq p_{j \leftarrow i}^t$ thus encodes directional influence on G_t .

4) *Scalar priority field from pairwise probabilities:* The pairwise probabilities $\{p_{i \leftarrow j}^t\}$ specify edge-level preferences between interacting agents, but do not directly yield a compact ordering over the whole interaction graph. We therefore assign each agent i at time t a scalar topological priority score $s_i^t \in \mathbb{R}$, where a larger value indicates that i tends to dominate (i.e., be yielded to by) its neighbors. The collection $\{s_i^t\}$ can be viewed as a scalar potential on the nodes of G_t whose pairwise differences summarize directed priority relations.

To derive supervision from these probabilities, we define an antisymmetric preference signal for each ordered pair:

$$A_{i \leftarrow j}^t = p_{j \leftarrow i}^t - p_{i \leftarrow j}^t, \quad A_{j \leftarrow i}^t = -A_{i \leftarrow j}^t. \quad (12)$$

With this convention, $A_{i \leftarrow j}^t > 0$ indicates that i should be assigned higher priority than j . We set $c_{i \leftarrow j}^t = |p_{i \leftarrow j}^t - \frac{1}{2}|^\alpha$ as the confidence of the corresponding preference, which down-weights uncertain pairs with $p_{i \leftarrow j}^t \approx \frac{1}{2}$. Let $E_t^+ = \{(j, i) \in E_t \mid c_{i \leftarrow j}^t > 0\}$ denote the positive-weight edge set, and let $\{\mathcal{C}_m^t\}_{m=1}^{M_t}$ be the connected components of the undirected support graph induced by E_t^+ . We then recover node scores by fitting score differences to these signed preferences via a weighted least-squares objective:

$$\{s_i^{t,*}\} = \arg \min_{\{s_i\}} \frac{1}{2} \sum_{(j,i) \in E_t^+} c_{i \leftarrow j}^t ((s_i - s_j) - A_{i \leftarrow j}^t)^2, \quad (13)$$

subject to the component-wise gauge constraints

$$\sum_{i \in \mathcal{C}_m^t} s_i = 0, \quad m = 1, \dots, M_t, \quad (14)$$

which remove the shift ambiguity independently in each connected component.

Equivalently, let $\mathbf{s}^t = [s_1, \dots, s_N]^\top$, let B^t be the incidence matrix of E_t^+ whose row for edge (j, i) has +1 at node i and -1 at node j , let $W^t = \text{diag}(c_{i \leftarrow j}^t)$, and let $\mathbf{A}^t$ stack the corresponding signed preferences $A_{i \leftarrow j}^t$. Then the objective can be written as

$$\frac{1}{2} \|(W^t)^{1/2} (B^t \mathbf{s}^t - \mathbf{A}^t)\|_2^2, \quad (15)$$

whose normal equations are

$$L^t \mathbf{s}^t = \mathbf{b}^t, \quad L^t = (B^t)^\top W^t B^t, \quad \mathbf{b}^t = (B^t)^\top W^t \mathbf{A}^t. \quad (16)$$

Here L^t is a weighted graph Laplacian. On each connected component, its null space is spanned only by the constant

vector; therefore, the component-wise zero-mean constraint removes this remaining degree of freedom and yields the unique zero-mean least-squares label

$$\mathbf{s}_{\mathcal{C}_m^t}^{t,*} = (L_{\mathcal{C}_m^t}^t)^\dagger \mathbf{b}_{\mathcal{C}_m^t}^t, \quad m = 1, \dots, M_t, \quad (17)$$

where $(\cdot)^\dagger$ denotes the Moore–Penrose pseudoinverse, and $L_{\mathcal{C}_m^t}^t$ and $\mathbf{b}_{\mathcal{C}_m^t}^t$ are the restrictions of L^t and $\mathbf{b}^t$ to component $\mathcal{C}_m^t$. Isolated nodes have no incident pairwise preference and are assigned the neutral label $s_i^{t,*} = 0$.

Solving this problem yields, for each time t , a scalar priority field $\{s_i^{t,*}\}$ that provides a component-wise graph-level ordering consistent (in the least-squares sense) with the edge-wise preferences. This least-squares procedure is used only to generate node-level supervision from centralized rollout data during training. During decentralized execution, each agent directly predicts $\hat{s}_i^t$ from its ego-centric local observation and does not construct or solve a global graph problem.

5) *Joint prediction and supervision of pairwise and scalar priorities:* At each time step t , agent i predicts edge-wise priority probabilities $\hat{p}_{i \leftarrow j}^t \in [0, 1]$ for each neighbor $j \in \mathcal{N}_i^t$ and a node-wise scalar priority score $\hat{s}_i^t \in \mathbb{R}$ for itself from its ego-centric observation o_i^t . These two quantities describe complementary levels of topology: $\hat{p}_{i \leftarrow j}^t$ encodes pairwise edge-level precedence, whereas $\hat{s}_i^t$ summarizes the node-level priority role of agent i in the local interaction graph. The training objective combines three terms. An edge-level loss fits the priorities,

$$\mathcal{L}_{\text{edge}} = \sum_t \sum_{(j,i) \in E_t} \text{BCE}(\hat{p}_{i \leftarrow j}^t, p_{i \leftarrow j}^t), \quad (18)$$

a node-level loss aligns the predicted scalar field with the least-squares solution,

$$\mathcal{L}_{\text{node}} = \sum_t \sum_i \|\hat{s}_i^t - s_i^{t,*}\|_2^2, \quad (19)$$

where $p_{i \leftarrow j}^t$ provides pairwise edge-level supervision, while $s_i^{t,*}$ provides a compact node-level priority label obtained by projecting the pairwise preferences onto a scalar priority field. The consistency term encourages the edge probabilities to agree with the ordering induced by the node scores. Specifically, we construct

$$\tilde{p}_{i \leftarrow j}^t = \sigma\left(\frac{\hat{s}_j^t - \hat{s}_i^t}{\tau_s}\right), \quad (20)$$

with temperature $\tau_s > 0$ and logistic sigmoid $\sigma(\cdot)$, and penalise their discrepancy via

$$\mathcal{L}_{\text{cons}} = \sum_t \sum_{(j,i) \in E_t} (\hat{p}_{i \leftarrow j}^t - \tilde{p}_{i \leftarrow j}^t)^2. \quad (21)$$

The overall topological loss is:

$$\mathcal{L}_{\text{topo}} = \mathcal{L}_{\text{edge}} + \lambda_{\text{node}} \mathcal{L}_{\text{node}} + \lambda_{\text{cons}} \mathcal{L}_{\text{cons}}, \quad (22)$$

with trade-off weights $\lambda_{\text{node}}, \lambda_{\text{cons}} > 0$. The consistency term ties the edge-level and node-level topology predictions. It encourages $\hat{p}_{i \leftarrow j}^t$ to be consistent with the ordering induced by $\hat{s}_j^t - \hat{s}_i^t$, thereby reducing contradictory priority estimates and improving the coherence of the learned topology representation for downstream coordination. Since the edge and node

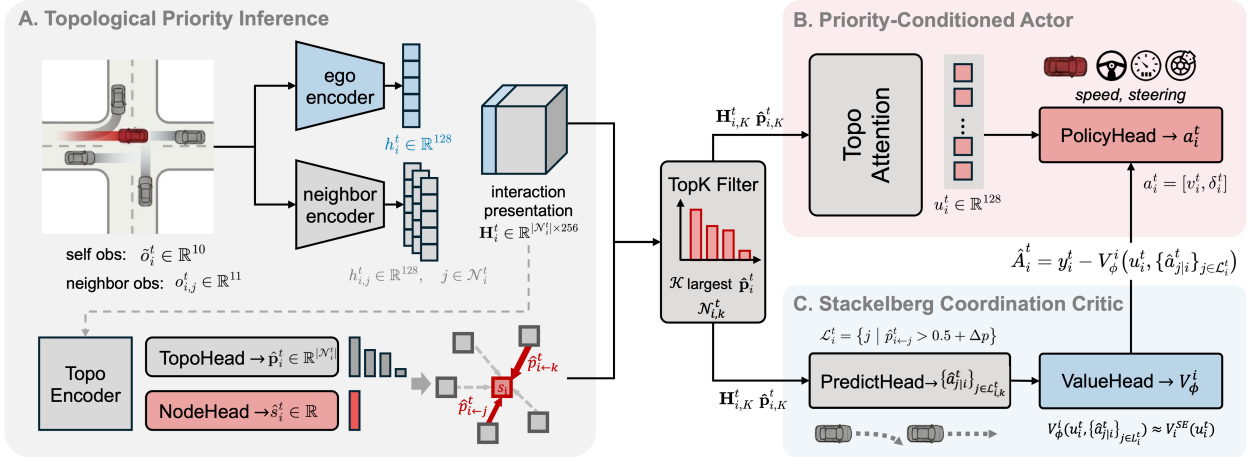


Fig. 2: **The TSC-Net architecture.** Ego and neighbor encoders construct an interaction representation $\mathbf{H}_i^t$, from which the TopoEncoder predicts pairwise priorities $\hat{\mathbf{p}}_{i \leftarrow j}^t$ and node scores $\hat{s}_i^t$ to form a local priority graph. Guided by this graph, a TopK filter and TopoAttention aggregate the most influential neighbors into a compact decision state u_i^t , which the PolicyHead maps to the continuous action a_i^t . The same priority structure defines the local leader set $\mathcal{L}_i^t$. The PredictHead estimates the actions of the selected neighbors $\{\hat{a}_{j|i}^t\}_{j \in \mathcal{N}_{i,K}^t}$, and the ValueHead conditions on the subset associated with the inferred leader set $\mathcal{L}_i^t$ to approximate the local Stackelberg value, yielding TD advantages $\hat{A}_i^t$ for actor optimization under CTDE.

predictions remain anchored by their corresponding labels in $\mathcal{L}_{\text{edge}}$ and $\mathcal{L}_{\text{node}}$, this term regularizes their agreement without replacing either supervision signal. At test time, each agent only requires its own observation o_i^t to compute $\hat{s}_i^t$ and $\hat{p}_{i \leftarrow j}^t$, enabling decentralized estimation of both pairwise and scalar priorities without any central coordinator or inter-vehicle communication.

B. Topology-conditioned Stackelberg Coordination Network (TSC-Net)

As shown in Fig. 2, we present the Topology-conditioned Stackelberg Coordination Network (TSC-Net), which instantiates TSC under centralized training and decentralized execution. Guided by the topological priority formulation in Sec. III-A, TSC-Net infers an ego-centric local priority graph from observations and uses it to enable topology-conditioned leader-follower coordination under decentralized, communication-free execution.

1) *Trajectory weaving for local priority graph inference:* Given the ego-centric observation o_i^t defined in Sec. II-A, agent i constructs an agent-centric representation for local priority graph inference. We encode the ego part and each neighbor part separately to obtain latent features:

$$h_i^t = \text{Encoder}_{\text{ego}}(\tilde{o}_i^t), \quad h_{i,j}^t = \text{Encoder}_{\text{nbr}}(o_{i,j}^t), \quad j \in \mathcal{N}_i^t, \quad (23)$$

where $h_i^t \in \mathbb{R}^{d_e}$ is the ego embedding, $h_{i,j}^t \in \mathbb{R}^{d_n}$ encodes the interaction from neighbor j as observed by agent i . Both encoders are implemented as MLPs, and $\text{Encoder}_{\text{nbr}}$ is shared across all neighbors so that the number of parameters is independent of $|\mathcal{N}_i^t|$. Concatenating the ego embedding with each neighbor embedding yields an ego-centric interaction representation

$$\mathbf{H}_i^t = \{[h_i^t, h_{i,j}^t]\}_{j \in \mathcal{N}_i^t} \in \mathbb{R}^{|\mathcal{N}_i^t| \times (d_e + d_n)}. \quad (24)$$

where $\mathbf{H}_i^t$ denotes the ego-neighbor interaction matrix over $\mathcal{N}_i^t$, whose rows are formed by concatenating the ego embedding with each neighbor embedding. Padding is used when the number of available neighbors is smaller than $|\mathcal{N}_i^t|$.

On top of $\mathbf{H}_i^t$ we introduce a topological priority encoding stream that recovers edge-wise and node-wise priorities induced by trajectory weaving:

$$\mathbf{Q}_i^{\text{topo}} = \text{TopoEncoder}(\mathbf{H}_i^t), \quad (25)$$

$$\hat{\mathbf{p}}_i^t = \text{TopoHead}(\mathbf{Q}_i^{\text{topo}}), \quad (26)$$

$$\hat{s}_i^t = \text{NodeHead}(\mathbf{Q}_i^{\text{topo}}). \quad (27)$$

Depicted in Fig. 2, the topological priority encoding stream takes the ego-neighbor interaction matrix $\mathbf{H}_i^t \in \mathbb{R}^{|\mathcal{N}_i^t| \times (d_e + d_n)}$ as input. A stack of neighbor-shared MLP blocks is employed as TopoEncoder to transform $\mathbf{H}_i^t$ into topology-aware latent representation $\mathbf{Q}_i^{\text{topo}} \in \mathbb{R}^{|\mathcal{N}_i^t| \times d_t}$, where each row corresponds to one ego-neighbor relation. Following Eqs. 26–27, two lightweight MLP heads are attached to these embeddings. TopoHead maps $\mathbf{Q}_i^{\text{topo}}$ to $|\mathcal{N}_i^t|$ edge-wise priority logits and applies an element-wise sigmoid activation to obtain $\hat{\mathbf{p}}_i^t \in [0, 1]^{|\mathcal{N}_i^t|}$, where each entry $\hat{p}_{i \leftarrow j}^t$ corresponds to the priority relation between agent i and neighbor $j \in \mathcal{N}_i^t$. In parallel, NodeHead performs masked pooling over $\mathbf{Q}_i^{\text{topo}}$ to obtain an ego-level topology embedding and predicts the scalar node-level priority score $\hat{s}_i^t \in \mathbb{R}$.

During training, the priority inference branch is supervised by the topological priority labels $p_{i \leftarrow j}^t$ and $s_i^{t,*}$. The edge labels satisfy the reciprocity constraint

$$p_{i \leftarrow j}^t + p_{j \leftarrow i}^t = 1. \quad (28)$$

The edge labels are de-cycled offline to suppress short directed loops in the induced priority graph. Consequently, each agent can infer a consistent local ordering signal from $\mathbf{H}_i^t$ under decentralized execution.

2) *Priority-Conditioned Local Decision Interface*: Before performing Stackelberg coordination, each agent compresses its local neighborhood into a compact, priority-filtered subset. The pairwise priorities $\hat{p}_{i \leftarrow j}^t$ provide a directional yielding cue under our arrow convention $i \leftarrow j$. Larger values indicate that neighbor j is more dominant in the inferred weaving relation and should therefore receive greater attention when agent i plans. The scalar score $\hat{s}_i^t$ complements these edge-level cues by providing an ego-level priority state within the local interaction graph.

To keep the per-agent control input bounded while retaining the most influential interactions, agent i selects the K neighbors with the largest priorities and aggregates their latent features:

$$\mathcal{N}_{i,K}^t = \text{Top} - K_K(\{\hat{p}_{i \leftarrow j}^t\}_{j \in \mathcal{N}_i^t}), \quad (29)$$

$$C_i^t = \text{TopoAttn}(h_i^t, \mathbf{H}_{i,K}^t, \hat{\mathbf{p}}_{i,K}^t) \in \mathbb{R}^{d_c}, \quad (30)$$

where $\mathcal{N}_{i,K}^t \subseteq \mathcal{N}_i^t$ is the selected neighbor set, $\mathbf{H}_{i,K}^t = \{h_{i,j}^t\}_{j \in \mathcal{N}_{i,K}^t}$ collects the priority-filtered neighbor embeddings, and $\hat{\mathbf{p}}_{i,K}^t = [\hat{p}_{i \leftarrow j}^t]_{j \in \mathcal{N}_{i,K}^t} \in [0, 1]^K$ is the corresponding priority vector. This priority-guided compression focuses the policy on strongly coupled interactions and supports scalability as traffic density increases. TopoAttention follows a cross-attention form over the selected local neighborhood. The ego embedding h_i^t is used as the query, while the selected neighbor embeddings $\{h_{i,j}^t\}_{j \in \mathcal{N}_{i,K}^t}$ are used as keys and values. Conditioned on the selected priorities, the attended aggregation yields the context representation C_i^t , which summarizes the interaction features most relevant to agent i for subsequent actor decoding.

On top of the aggregated context, we define the actor branch of TSC-Net. The attended context C_i^t is already conditioned on the selected pairwise priorities $\hat{\mathbf{p}}_{i,K}^t$, while $\hat{s}_i^t$ further provides an ego-level priority state for actor decoding. These topology-conditioned cues are mapped to a local decision state u_i^t , which parametrizes a stochastic policy:

$$u_i^t = \text{EgoDecoder}(C_i^t, \hat{s}_i^t), \quad (31)$$

$$a_i^t \sim \pi_\theta(\cdot | u_i^t) = \text{PolicyHead}_\theta(u_i^t). \quad (32)$$

The PolicyHead outputs the mean and diagonal standard deviation of a Gaussian distribution. The sampled action is then squashed by tanh and rescaled to the continuous speed-steering control space.

This subsection specifies the actor interface only. The critic is introduced in Sec. III-B3, where it conditions on u_i^t together with predicted actions of inferred local leaders to evaluate the priority-guided leader-follower structure, while keeping such predictions out of the policy input for training stability.

3) *Stackelberg Coordination on the Topological Priority Graph*: We define a Stackelberg-style local decision rule on the inferred priority graph by identifying, for each agent i , a subset of neighbors that are treated as *local leaders* at time t . Among the Top- K neighbors $\mathcal{N}_{i,K}^t$ selected in (29), we define:

$$\mathcal{L}_i^t = \left\{ j \in \mathcal{N}_{i,K}^t \mid \hat{p}_{i \leftarrow j}^t > \frac{1}{2} + \Delta_p \right\}, \quad (33)$$

where $\Delta_p \geq 0$ is a small margin to avoid unstable ties around 0.5, i.e., the neutral threshold at which neither agent has a confident precedence over the other. We set $\Delta_p = 0.05$ in all experiments. Applied after (29), this margin suppresses ambiguous priorities near 0.5 and keeps the leader set bounded by K . This prevents weak priority differences from causing frequent changes in the inferred leader set. Under the arrow convention $i \leftarrow j$, $\mathcal{L}_i^t$ collects those neighbors whose inferred weaving relation suggests that i should yield to j . If $\mathcal{L}_i^t = \emptyset$, the critic uses no leader action conditioning.

Under parameter sharing, agents execute different instances of the same policy π_θ . However, the realized leader actions $a_j^t \sim \pi_\theta(\cdot | u_j^t)$ are not observable to agent i at decision time. We therefore predict one-step neighbor actions using the priority-filtered embeddings $\mathbf{H}_{i,K}^t = \{h_{i,j}^t\}_{j \in \mathcal{N}_{i,K}^t}$ and priorities $\hat{\mathbf{p}}_{i,K}^t = \{\hat{p}_{i \leftarrow j}^t\}_{j \in \mathcal{N}_{i,K}^t}$

$$\hat{\mathbf{a}}_i^t = \text{PredictHead}(\mathbf{H}_{i,K}^t, \hat{\mathbf{p}}_{i,K}^t), \quad (34)$$

where $\hat{\mathbf{a}}_i^t = \{\hat{a}_{j|i}^t\}_{j \in \mathcal{N}_{i,K}^t}$ are the predicted neighbor actions as perceived by agent i .

The prediction head outputs one continuous action vector for each selected neighbor and is supervised by the neighbor action recorded in the sampled future rollout. For critic evaluation under decentralized information, the true leader actions are unavailable to agent i and are therefore replaced by the predicted actions. We use $\{\hat{a}_{j|i}^t\}_{j \in \mathcal{L}_i^t}$ as topology-guided approximations of the (unobserved) leader actions in the local coordination game.

Given the ego decision state u_i^t and predicted leader actions, we implement a Stackelberg-conditioned critic via

$$V_\phi^i(u_i^t, \{\hat{a}_{j|i}^t\}_{j \in \mathcal{L}_i^t}) = \text{ValueHead}(u_i^t, \{\hat{a}_{j|i}^t\}_{j \in \mathcal{L}_i^t}), \quad (35)$$

trained by temporal-difference learning to estimate the expected discounted return of agent i under the induced local hierarchy. The value head receives the ego decision representation and the predicted actions of selected leaders.

We formalize this target through the local Stackelberg state-value $V_i^{\text{SE}}(u_i^t)$ and enforce

$$V_\phi^i(u_i^t, \{\hat{a}_{j|i}^t\}_{j \in \mathcal{L}_i^t}) \approx V_i^{\text{SE}}(u_i^t). \quad (36)$$

This construction can be interpreted through classical Stackelberg games, where a leader commits to a policy and a follower responds optimally. While the standard equilibrium is defined by the bilevel program:

$$\pi_L^* = \arg \max_{\pi_L} J_L(\pi_L, \text{BR}(\pi_L)), \quad \pi_F^* = \text{BR}(\pi_L^*), \quad (37)$$

Our objective is not to solve (37) globally, but to realize follower-side evaluation locally on the priority graph. The priority graph provides a local time-varying ordering signal. Neighbors in $\mathcal{L}_i^t$ are treated as temporarily committed leaders, and agent i evaluates its action as a follower response to them. For agent i , treating $\mathcal{L}_i^t$ as leaders yields the Bellman recursion:

$$V_i^{\text{SE}}(u_i^t) = \mathbb{E} \left[r_i^t + \gamma V_i^{\text{SE}}(u_i^{t+1}) \mid u_i^t, a_i^t \sim \pi_\theta(\cdot | u_i^t), a_j^t \sim \pi_\theta(\cdot | u_j^t), \forall j \in \mathcal{L}_i^t \right], \quad (38)$$

where neighbors in $\mathcal{L}_i^t$ act as local leaders for i under parameter sharing. Maximizing $J_i^{\text{SE}}(\pi_\theta) = \mathbb{E}[V_i^{\text{SE}}(u_i^0)]$ thus corresponds to seeking a local best response to the leader set:

$$\text{BR}_i(\{\pi_\theta\}_{j \in \mathcal{L}_i^t}) \in \arg \max_{\pi_\theta} J_i^{\text{SE}}(\pi_\theta). \quad (39)$$

During critic training, the leader actions in (38) are unavailable from agent i 's local information and are therefore replaced by predictions $\hat{a}_{j|i}^t$. Conditioning V_ϕ^i on $\{\hat{a}_{j|i}^t\}_{j \in \mathcal{L}_i^t}$ therefore yields a biased but tractable approximation to the local Stackelberg backup. Since predicted leader actions are used only by the critic, the actor remains conditioned solely on u_i^t . Under mild assumptions on the priority graph and bounded prediction error, the induced value bias remains controlled and the resulting advantage estimates support stable policy-gradient updates. Lemma 1 and Lemma 2 in Appendix provide a formal statement and connect the induced update to the underlying local bilevel structure.

Crucially, the actor depends only on u_i^t and does not take $\{\hat{a}_{j|i}^t\}$ as direct inputs. Predicted leader actions shape the value landscape through the critic, rather than being hard-coded into the policy parameterization, which promotes stable policy optimization while preserving topology-conditioned coordination. Thus, leader-action prediction is used only for critic-side training, while decentralized execution uses only the actor with locally available inputs.

C. Learning Objective and Training Procedure

We train TSC-Net under the CTDE paradigm. The overall objective consists of (i) supervised learning of the topological priority graph, (ii) neighbor-action prediction loss, and (iii) actor-critic objective for topology-conditioned control.

The priority inference branch (Sec. III-B1) predicts edge probabilities $\hat{p}_{i \leftarrow j}^t$ and node scores $\hat{s}_i^t$ from o_i^t and is supervised by the trajectory-derived labels $p_{i \leftarrow j}^{t,*}$ and $s_i^{t,*}$. We adopt the priority inference loss $\mathcal{L}_{\text{topo}}$ defined in Eq. 22 to train this branch.

The prediction head outputs $\hat{a}_{j|i}^t$ and is trained on the selected Top- K neighbors $j \in \mathcal{N}_{i,K}^t$ with

$$\mathcal{L}_{\text{lead}} = \sum_{t,i} \sum_{j \in \mathcal{N}_{i,K}^t} \ell_{\text{act}}(\hat{a}_{j|i}^t, a_j^t), \quad (40)$$

where ℓ_{act} is an MSE regression loss. The target action a_j^t is the realized neighbor action recorded in the sampled future rollout during centralized training. The Stackelberg-conditioned critic uses only the predictions associated with the leader set $\mathcal{L}_i^t$.

Given rewards r_i^t and discount γ , we train the Stackelberg-conditioned critic with a squared temporal-difference loss:

$$\mathcal{L}_{\text{value}} = \sum_{t,i} \left(V_\phi^i(u_i^t, \{\hat{a}_{j|i}^t\}_{j \in \mathcal{L}_i^t}) - y_i^t \right)^2, \quad (41)$$

where y_i^t is a one-step bootstrapped target computed using a target critic. The actor is updated with an advantage-weighted log-likelihood objective:

$$\mathcal{L}_{\text{policy}} = - \sum_{t,i} \log \pi_\theta(a_i^t | u_i^t) \hat{A}_i^t, \quad (42)$$

with detached advantages:

$$\hat{A}_i^t = y_i^t - V_\phi^i(u_i^t, \{\hat{a}_{j|i}^t\}_{j \in \mathcal{L}_i^t}). \quad (43)$$

We minimize the aggregate objective:

$$\mathcal{L} = \mathcal{L}_{\text{policy}} + \lambda_V \mathcal{L}_{\text{value}} + \lambda_{\text{topo}} \mathcal{L}_{\text{topo}} + \lambda_{\text{lead}} \mathcal{L}_{\text{lead}}, \quad (44)$$

with non-negative weights $\lambda_V, \lambda_{\text{topo}}, \lambda_{\text{lead}}$. All loss weights and training hyperparameters are summarized in Table II.

Training proceeds by collecting joint trajectories with the current decentralized policy. Mini-batches sampled from the future rollouts are used to minimize $\mathcal{L}$, jointly updating the priority inference modules and the topology-conditioned actor-critic. All agents share network parameters during training and execute the same decentralized policy using local decision states.

IV. EXPERIMENTS

A. Evaluation Platform

We build our multi-agent traffic environment on VMAS [43] and follow the environment design and evaluation protocol of SigmaRL [9]. The simulator enables efficient parallel rollouts with low-level continuous control for all vehicles. Fig. 3 shows representative maps. We evaluate on four traffic scenarios:

Clover: A cloverleaf interchange with repeated merge-diverge segments and frequent short-range conflicts (initialize 20 vehicles).

Weave: A compact skewed intersection with tightly coupled turning and crossing flows (initialize 8 vehicles).

Merge: An on-ramp merging task requiring gap acceptance and cooperative speed adjustment (initialize 8 vehicles).

Bypass: A looped bypass connecting parallel road segments via an alternative curved path (initialize 8 vehicles).

We report three evaluation metrics: collision rate (CR), average speed (AS), and smoothness (SM). CR is decomposed into agent-agent and agent-map events. Let T be the episode length and $\mathbb{I}[\cdot]$ the indicator:

$$\begin{aligned} \text{CR}_{\text{AA}} &\triangleq \frac{1}{T} \sum_{t=1}^T \mathbb{I}[\exists i : \text{collide}_{i,\text{agent}}^t] \times 100, \\ \text{CR}_{\text{AM}} &\triangleq \frac{1}{T} \sum_{t=1}^T \mathbb{I}[\exists i : \text{collide}_{i,\text{map}}^t] \times 100, \end{aligned} \quad (45)$$

and $\text{CR} = \text{CR}_{\text{AA}} + \text{CR}_{\text{AM}}$.

AS measures traffic efficiency as the mean speed magnitude over agents and time steps, normalized by v_{max} :

$$\text{AS} \triangleq \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T \frac{\|v_i^t\|}{v_{\text{max}}} \times 100. \quad (46)$$

SM measures control variability via step-to-step changes in longitudinal and steering commands:

$$\begin{aligned} \text{SM} &\triangleq \frac{1}{N(T-1)} \sum_{i=1}^N \sum_{t=2}^T \left[\beta \frac{|v_i^t - v_i^{t-1}|}{2v_{\text{max}}} \right. \\ &\quad \left. + (1-\beta) \frac{|\delta_i^t - \delta_i^{t-1}|}{2\delta_{\text{max}}} \right] \times 100. \end{aligned} \quad (47)$$

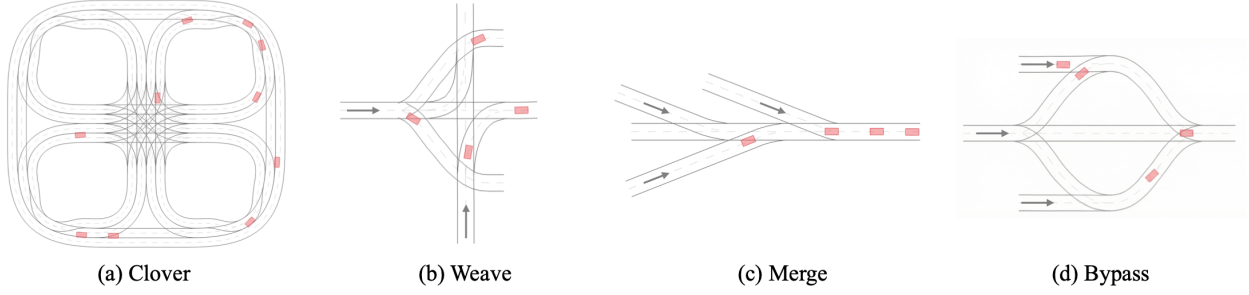


Fig. 3: Evaluation scenarios for dense multi-vehicle interaction.

TABLE I: Testing results on four scenarios from different methods. Collision rate (CR[%]) measures safety and is decomposed into agent-agent collisions CR_{AA} and agent-map collisions CR_{AM} . Average speed (AS[%]) measures traffic efficiency. Smoothness (SM[%]) measures control stability. Results are reported as mean $\pm$ standard deviation over five random seeds. Best and second-best mean values are bolded and underlined within each scenario.

		FIFO	MFPO	SigmaRL	XP-MARL	TSC
Clover	$CR_{AA} \downarrow$	1.78 ± 0.21	1.71 ± 0.24	1.67 ± 0.27	0.19 ± 0.10	0.04 ± 0.04
	$CR_{AM} \downarrow$	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	<u>0.03 ± 0.02</u>	<u>0.03 ± 0.02</u>
	$CR \downarrow$	1.78 ± 0.21	1.71 ± 0.24	1.67 ± 0.27	<u>0.22 ± 0.09</u>	0.07 ± 0.03
	$AS \uparrow$	71.7 ± 0.88	86.6 ± 0.84	<u>87.1 ± 0.82</u>	84.8 ± 0.90	88.2 ± 0.65
	$SM_{LO} \downarrow$	5.17 ± 0.63	2.63 ± 0.44	2.58 ± 0.48	<u>2.27 ± 0.52</u>	1.95 ± 0.57
	$SM_{LA} \downarrow$	15.5 ± 0.40	11.2 ± 0.32	10.4 ± 0.45	<u>8.95 ± 0.37</u>	8.89 ± 0.33
	$SM \downarrow$	10.3 ± 0.52	6.92 ± 0.36	6.50 ± 0.35	<u>5.61 ± 0.42</u>	5.42 ± 0.39
Weave	$CR_{AA} \downarrow$	6.56 ± 0.35	5.46 ± 0.27	5.45 ± 0.50	<u>0.69 ± 0.39</u>	0.18 ± 0.21
	$CR_{AM} \downarrow$	0.00 ± 0.00	2.29 ± 0.13	0.95 ± 0.24	<u>0.98 ± 0.04</u>	<u>0.81 ± 0.22</u>
	$CR \downarrow$	6.56 ± 0.35	7.75 ± 0.21	6.40 ± 0.51	<u>1.66 ± 0.40</u>	0.99 ± 0.28
	$AS \uparrow$	64.2 ± 0.67	72.5 ± 0.37	<u>79.2 ± 0.67</u>	78.2 ± 0.55	81.8 ± 0.41
	$SM_{LO} \downarrow$	13.4 ± 0.49	9.44 ± 0.15	7.59 ± 0.21	<u>5.58 ± 0.39</u>	5.29 ± 0.34
	$SM_{LA} \downarrow$	14.2 ± 0.20	16.5 ± 0.16	14.9 ± 0.20	<u>9.51 ± 0.20</u>	9.38 ± 0.15
	$SM \downarrow$	13.8 ± 0.35	13.0 ± 0.10	11.3 ± 0.15	<u>7.54 ± 0.25</u>	7.33 ± 0.19
Merge	$CR_{AA} \downarrow$	3.83 ± 0.02	4.37 ± 0.19	4.28 ± 0.39	<u>0.24 ± 0.29</u>	0.03 ± 0.01
	$CR_{AM} \downarrow$	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
	$CR \downarrow$	3.83 ± 0.02	4.37 ± 0.19	4.28 ± 0.39	<u>0.24 ± 0.29</u>	0.03 ± 0.01
	$AS \uparrow$	71.4 ± 0.79	78.6 ± 0.81	<u>83.1 ± 0.85</u>	81.4 ± 0.29	84.7 ± 0.56
	$SM_{LO} \downarrow$	7.72 ± 0.43	5.85 ± 0.24	<u>5.23 ± 0.38</u>	5.57 ± 0.29	5.09 ± 0.42
	$SM_{LA} \downarrow$	12.3 ± 0.32	10.0 ± 0.30	<u>8.56 ± 0.10</u>	5.32 ± 0.15	<u>6.58 ± 0.15</u>
	$SM \downarrow$	10.0 ± 0.35	7.95 ± 0.28	6.90 ± 0.23	5.45 ± 0.21	<u>5.84 ± 0.27</u>
Bypass	$CR_{AA} \downarrow$	6.91 ± 0.38	4.80 ± 0.36	5.02 ± 0.26	<u>0.56 ± 0.22</u>	0.08 ± 0.13
	$CR_{AM} \downarrow$	0.00 ± 0.00	0.78 ± 0.22	<u>0.03 ± 0.04</u>	0.07 ± 0.06	<u>0.03 ± 0.03</u>
	$CR \downarrow$	6.91 ± 0.38	5.57 ± 0.34	5.06 ± 0.26	<u>0.62 ± 0.21</u>	0.10 ± 0.15
	$AS \uparrow$	68.6 ± 0.79	76.4 ± 0.43	81.4 ± 0.96	78.0 ± 0.73	<u>79.4 ± 0.56</u>
	$SM_{LO} \downarrow$	9.29 ± 0.48	6.93 ± 0.20	<u>6.18 ± 0.42</u>	5.94 ± 0.56	7.44 ± 0.42
	$SM_{LA} \downarrow$	10.2 ± 0.36	14.5 ± 0.44	14.4 ± 0.17	9.02 ± 0.18	<u>9.16 ± 0.15</u>
	$SM \downarrow$	9.75 ± 0.37	10.7 ± 0.32	10.3 ± 0.20	7.48 ± 0.34	<u>8.30 ± 0.27</u>

where the first and second terms define the longitudinal and lateral components SM_{LO} and SM_{LA} , respectively. Here we set $\beta = 0.5$.

B. Experiment Setting

We compare TSC with four baselines: FIFO, MFPO, SigmaRL, and XP-MARL. FIFO is a rule-based right-of-way baseline based on first-in-first-out and arrival-order scheduling in intersection management [48], [49]. It assigns higher priority to vehicles that are estimated to reach the nearest conflict region earlier, while later vehicles yield by decelerating or

waiting. MFPO uses a mean-field aggregation of nearby vehicles instead of explicit pairwise interactions, enabling scalable learning and execution as the number of agents grows [50], [51]. SigmaRL is a MARL approach that emphasizes sample-efficient and generalizable value learning.

XP-MARL extends SigmaRL with auxiliary prioritization and centralized sequential coordination, where a central controller establishes the priority order and propagates higher-priority actions [23]. TSC addresses the same coordination problem through graph-local leader-follower relations inferred from local observations, without constructing a global order

TABLE II: Training hyperparameters of TSC-Net.

Category	Parameter	Value
Training Setup	Optimizer	Adam
	Learning rate (policy)	2×10^{-4}
	Learning rate (action predictor)	3×10^{-4}
	Discount factor γ	0.99
	Training iterations	250
	Frames per iteration	4096
	Parallel environments	32
	Minibatch size	512
Loss Weights (Eq. 22, 44)	Max gradient norm	1.0
	Value loss weight λ_V	1.0
	Topology loss weight λ_{topo}	0.5
	Leader prediction weight λ_{lead}	1.0
	Node score weight λ_{node}	1.0
	Consistency weight λ_{cons}	1.0
Topology Parameters	Softmax temperature τ	1.0
	Stabilizing constant ε	1×10^{-3}
	Confidence exponent α	1.0
	Consistency temperature τ_s	1.0
	Leader margin Δ_p	0.05
	Future-trajectory horizon H	5 steps
Neighborhood	Inference neighborhood $ \mathcal{N}_i^t $	4
	Execution neighborhood K	2

or exchanging inter-agent actions. Thus, XP-MARL provides the closest sequential-coordination reference, although the two methods differ in their execution information structures. Together with MFPO and SigmaRL, these methods represent complementary interaction models for continuous multi-vehicle coordination.

The learning-based comparison focuses on task-oriented methods for continuous multi-vehicle coordination under a common environment and control interface. TSC contributes a topology-conditioned coordination mechanism built on a CTDE actor-critic framework, and this mechanism is not tied to a specific policy-optimization backbone. QMIX would require discretizing the continuous speed-steering action space, while evaluations with alternative actor-critic backbones such as MADDPG and MAPPO are left for future work.

All learning-based methods are trained in the *Clover* scenario, and all methods, including FIFO, are subsequently evaluated in all four scenarios under consistent environment settings. For a controlled comparison, all learning-based methods use the same environment dynamics, observation design, continuous action space, reward function, episode horizon, and evaluation protocol.

Each episode comprises 1200 steps with $\Delta t = 0.05$ s. Under partial observability, we set $|\mathcal{N}_i^t| = 4$ for priority inference and $|\mathcal{N}_{i,K}^t| = 2$ for policy learning and execution.

Each learning-based method is independently trained using five random seeds, and all methods are evaluated over five independently seeded runs, with results reported as mean $\pm$ standard deviation. All experiments are conducted on an AMD Ryzen 9 7945HX CPU with 32 GB RAM.

Under the configuration specified in Table II, training each model takes less than one hour. We also measure the average single-step policy inference time on the same hardware. The measured latency remains below the 50 ms control interval, supporting real-time decentralized execution.

C. Main Results

Table I summarizes performance on four scenarios. TSC achieves the lowest total collision rate in all scenarios while maintaining competitive average speed, which indicates that the safety improvement is not obtained by conservative slow-downs. The gain is most pronounced in *Weave*, where TSC attains a collision rate of 0.99% with the highest AS of 81.8. In comparison, FIFO, MFPO, and SigmaRL exhibit substantially higher collision rates in the same scenario, at 6.56%, 7.75%, and 6.40%, respectively. Similar safety improvements appear in *Clover*, *Merge*, and *Bypass*, where TSC consistently reduces collisions to near-zero levels and preserves traffic efficiency.

Decomposed results show that the reduction is mainly driven by fewer agent-agent collisions. This is consistent with our design that infers a directed local priority graph and uses it to structure interactions under decentralized execution. For example, in *Weave* the agent-agent collision rate drops to 0.18% under TSC, compared with 5.46% under MFPO and 5.45% under SigmaRL. The higher collision rate of FIFO further indicates that a fixed arrival-order rule is insufficient for dynamically coupled interactions. XP-MARL also improves safety through sequential coordination, with centralized priority assignment and higher-priority action propagation constituting its core coordination mechanism. In contrast, TSC attains lower total collision rates across all scenarios without centralized priority management or inter-agent communication. This result indicates that topology-conditioned graph-local coordination can provide an effective alternative to centralized global sequencing in dense traffic. TSC also achieves the best or second best smoothness in most cases, with the main exceptions on *Merge* and *Bypass*, where XP-MARL yields smoother control on SM_{LA} or SM_{LO} .

Fig. 4 presents radar plots over safety, average speed, and smoothness. For each scenario, we normalize each metric to its maximal and minimal values over all evaluated episodes and all learning-based methods, and use the complement of the normalized collision rate as the safety score, so that larger values consistently represent safer performance. The plots show that XP-MARL and TSC both expand the safety axis, highlighting the benefit of sequential Stackelberg-style coordination, whereas TSC yields a more balanced profile across all axes due to topology-guided sparsification and graph-local leader-follower reasoning, which stabilizes interactions and avoids over-conditioning on centralized action propagation.

To further evaluate the generalization under varying traffic density, we execute the learned policies in the same test scenarios while changing the number of vehicles. As shown in Fig. 5, MFPO and SigmaRL are more sensitive to the number of vehicles, with collision rates increasing rapidly as the traffic becomes denser. In contrast, XP-MARL and TSC maintain substantially lower collision rates across all tested densities. Notably, TSC consistently yields the lowest collision rates and exhibits the slowest degradation as the number of vehicles increases, indicating that the topology-conditioned Stackelberg structure supports scalable and stable coordination by converting dense interactions into sparse leader-follower dependencies without constructing a centralized global order.

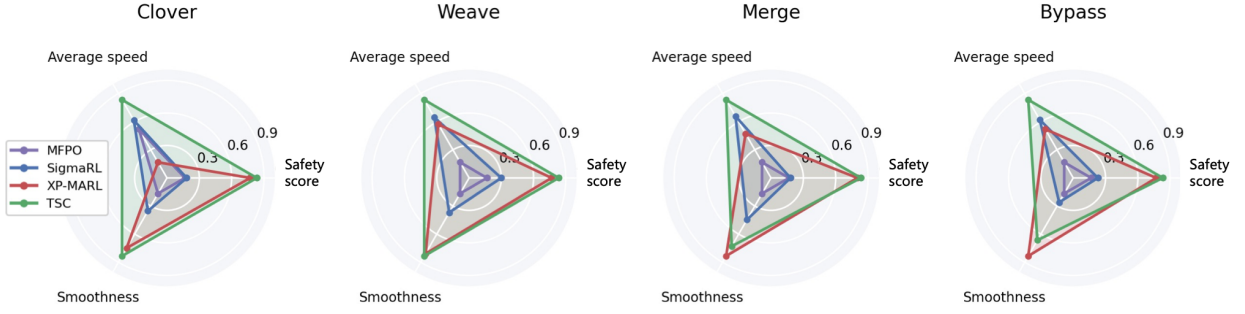


Fig. 4: Comparison of safety, efficiency, and smoothness across four scenarios.

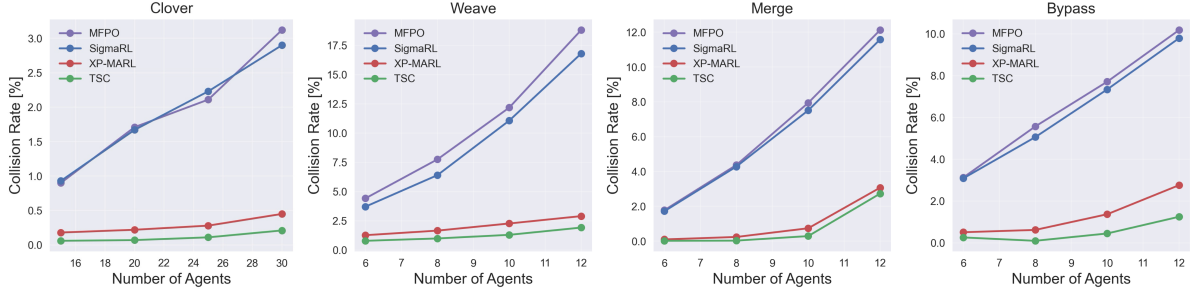


Fig. 5: Collision rate of different methods under varying numbers of vehicles at test time.

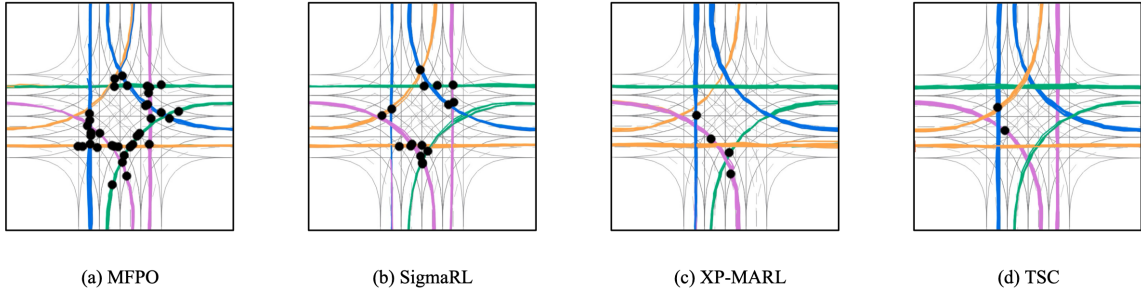


Fig. 6: Visualization of vehicle trajectories in one test episode. Black dots indicate collision events.

Comparative training curves over five independent random seeds are provided in Appendix C.1, while quantitative evaluations of leader-action prediction accuracy are reported in Appendix C.3.

D. Behavior Analysis

In Fig. 6, MFPO and SigmaRL show dense trajectory crossings and collision clusters around the conflict region, which indicates unresolved right-of-way interactions under decentralized control. XP-MARL reduces collisions by enforcing sequential reactions, yet residual collision markers remain near merging and crossing areas. TSC produces clearer precedence patterns with better separated traffic streams and only sparse collision events, illustrating how topology-conditioned priority inference can stabilize local leader-follower responses without communication.

Across four scenarios, Fig. 7 highlights recurrent interaction patterns induced by TSC, including yielding and queuing near conflict points, and proactive gap creation behaviors such as cutting in, bypassing, and occasional reversing. These

behaviors arise from local observations with a shared policy, showing that the inferred priority graph provides a compact coordination signal that transfers across different geometries and traffic densities.

E. Ablation Studies

We conduct ablation studies to assess the contribution of each component in the proposed coordination framework. Table III compares TSC with five variants, with all results reported as mean $\pm$ standard deviation over five random seeds. “TSC with random priority,” “TSC without Stackelberg,” and “TSC without Top- K filter” yield average collision rates of 1.21%, 1.01%, and 1.69%, respectively, showing substantial degradation when priority inference, leader-follower coordination, or sparse neighbor selection is removed. “TSC with Euclidean Top- K ” uses the same K size as TSC but selects neighbors according to Euclidean distance. Its average collision rate increases to 0.61%, compared with 0.30% for TSC. Although the two selection strategies often retain similar neighbors, Euclidean distance cannot distinguish nearby

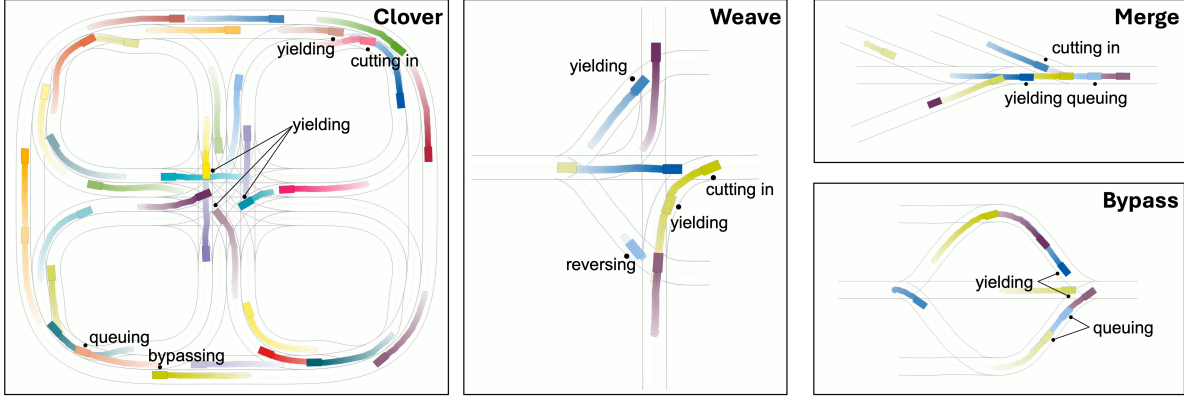


Fig. 7: Qualitative trajectory visualization of TSC across four interaction scenarios, highlighting emergent behaviors including yielding, queuing, cutting in, bypassing, and reversing. For each vehicle, we plot a 25-step trajectory segment with distinct colors, where the luminosity fades from light to deep to indicate earlier to later positions.

parallel or following vehicles from more critical vehicles approaching in opposing or crossing directions. “TSC without $\mathcal{L}_{\text{cons}}$ ” also shows moderate degradation, with an average collision rate of 0.65%, supporting the role of consistency regularization in aligning edge-wise and node-wise priority estimates. Since all variants share the same communication-free execution interface, the ablations isolate the effects of TSC’s internal components. These effects are complementary rather than additive, as the components are jointly optimized during policy learning. The corresponding training curves are provided in Appendix C.2, further illustrating how the individual components affect collision reduction, reward improvement, and convergence behavior during policy learning.

Fig. 8 further studies the influence of neighborhood size for sparse selection on agent-agent collision rate. We compare TopoNbr-1/2/3/5, where the policy conditions on the topologically most relevant K neighbors. As illustrated in Fig. 8, TopoNbr-2 achieves the lowest collision rate and the fastest reduction during training, indicating improved stability and sample efficiency. TopoNbr-1 tends to miss secondary conflict partners, while larger K includes weakly coupled neighbors, which dilutes the influence of the truly critical vehicles and injects inconsistent interaction cues, resulting in slower and less consistent convergence and more agent-agent collisions. These results indicate that retaining a compact set of interaction-critical neighbors provides a favorable balance between information coverage and interaction relevance.

Table IV evaluates the sensitivity of TSC to the leader-selection margin Δp and the stabilizing constant ϵ . Setting $\Delta p = 0$ increases the average collision rate from 0.43% to 1.20%, whereas $\Delta p = 0.05$ achieves the lowest collision rate. Increasing the margin to 0.20 degrades performance to 0.97%, indicating that a moderate margin effectively filters ambiguous priority estimates near 0.5 without suppressing high-confidence leader relations. For ϵ , the average collision rate varies only from 0.41% to 0.50% over the range 10^{-4} – 10^{-2} . The default setting $\epsilon = 10^{-3}$ yields 0.43%, close to the best result of 0.41% obtained at 5×10^{-4} . These results demonstrate that the weaving-distance formulation is robust to

the choice of ϵ within the evaluated range.

Overall, the ablation results demonstrate the complementary roles of topological priority inference, Stackelberg-style value conditioning, and compact neighborhood selection in improving coordination safety. They further support the use of a moderate leader-selection margin and confirm the robustness of the weaving-distance formulation to its stabilizing constant.

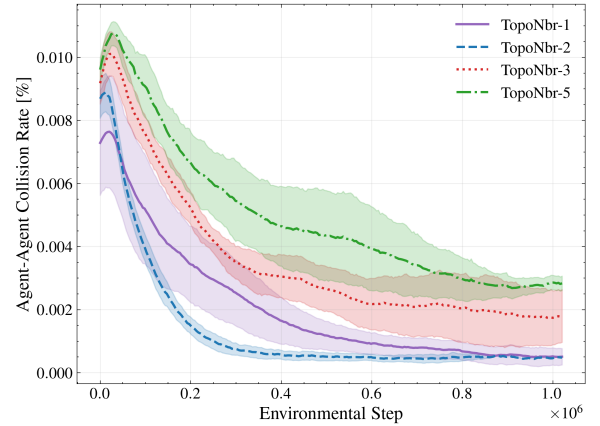


Fig. 8: Comparison of TSC with different TopoNbr- K settings.

TABLE III: Ablation study of the total collision rate (CR [%]) across four scenarios, reported as mean $\pm$ standard deviation over five random seeds.

Method	Clover	Weave	Merge	Bypass
w/ random priority	0.30 $\pm$ 0.20	1.62 $\pm$ 0.32	1.11 $\pm$ 0.16	1.79 $\pm$ 0.44
w/o Stackelberg	0.27 $\pm$ 0.20	1.08 $\pm$ 0.58	0.78 $\pm$ 0.43	1.91 $\pm$ 0.63
w/o Top- K filter	1.00 $\pm$ 0.26	2.92 $\pm$ 0.62	0.97 $\pm$ 0.24	1.87 $\pm$ 0.48
w/ Euclidean Top- K	0.19 $\pm$ 0.04	1.35 $\pm$ 0.18	0.19 $\pm$ 0.07	0.71 $\pm$ 0.29
w/o $\mathcal{L}_{\text{cons}}$	0.26 $\pm$ 0.11	1.47 $\pm$ 0.29	0.23 $\pm$ 0.24	0.64 $\pm$ 0.37
TSC	0.07 $\pm$ 0.03	0.99 $\pm$ 0.28	0.03 $\pm$ 0.01	0.10 $\pm$ 0.15

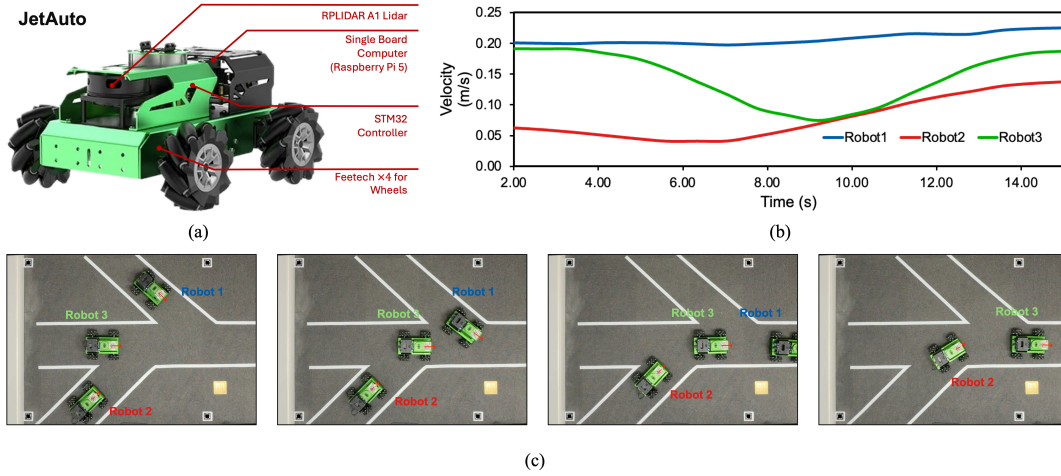


Fig. 9: Real-world proof-of-concept experiment using three JetAuto mobile robots. (a) Hardware overview of the JetAuto platform. (b) Measured velocity profiles during the three-robot merge interaction. (c) Representative overhead snapshots showing that the learned decentralized policy induces a clear yielding-passing sequence during conflict resolution.

TABLE IV: Sensitivity analysis of leader-selection margin Δp and stabilizing constant ϵ on average collision rate (Avg. CR[%]) across four scenarios and five random seeds. The default setting is $\epsilon = 10^{-3}$ and $\Delta p = 0.05$.

Parameter	Candidate values				
Δp	0	0.02	0.05	0.10	0.20
Avg. CR ↓	1.20	0.72	0.43	0.46	0.97
ϵ	10^{-4}	5×10^{-4}	10^{-3}	10^{-2}	
Avg. CR ↓	0.50	0.41	0.43	0.47	

F. Real-World Proof-of-Concept Experiment

We conduct a proof-of-concept evaluation of TSC using three JetAuto mobile robots¹ in an indoor testbed configured to reproduce the *Merge* interaction scenario, as shown in Fig. 9. Each JetAuto is a four-wheel mobile platform equipped with Mecanum wheels and an onboard controller for low-level motion execution. Lateral motion is disabled to approximate the car-like kinematics considered during training. The experimental system further includes a fixed overhead camera and an external computer for visual processing and policy inference. The camera provides real-time planar poses of the robots within the experimental workspace.

At each control step, the simulation-side measurement interface used to construct local observations is replaced by camera-based pose measurements from the physical testbed. These measurements are registered with the predefined lane geometry and reference paths to reconstruct the ego-centric observations specified in Appendix B.1.2. Although the sensing, motion, and actuation interfaces differ from those in simulation, the observation and action semantics remain unchanged: the reconstructed observations follow the training representation, and the normalized policy outputs are converted into longitudinal and angular velocity commands supported by the robots. Policy inference is performed on an external PC, which independently generates a command for each robot using only

its corresponding observation, without access to other robots' observations or actions and without inter-robot communication. These interface adaptations enable direct deployment without retraining or fine-tuning.

Fig. 9(c) presents representative snapshots of the three-robot merge interaction. As the robots approach the conflict region, each independently infers local leader-follower relations from its ego-centric observation. Robot 1 proceeds as the local leader, robot 3 decelerates to yield to robot 1 while acting as the leader of robot 2, and robot 2 maintains a lower velocity until the preceding robots clear the merge. The resulting local hierarchy follows the sequence robot 1 $\rightarrow$ robot 3 $\rightarrow$ robot 2. The velocity profiles in Fig. 9(b) support this interpretation, showing the initial deceleration of robot 3, the sustained yielding of robot 2, and their sequential recovery after the corresponding leaders pass. The absence of prolonged stopping or oscillatory yielding provides qualitative evidence of coordination efficiency, while the successful trial demonstrates deployment feasibility under the nominal sensing noise, actuation disturbances, and control latency of the testbed.

V. CONCLUSION

This paper introduced TSC, a topology-conditioned Stackelberg coordination framework for resolving spatiotemporal conflicts among decentralized autonomous agents in dense traffic. A weaving-induced directed priority graph encodes asymmetric local precedence, and its edges are interpreted as leader-follower constraints that couple local Stackelberg subgames without constructing a global order of play. Based on this formulation, TSC-Net infers ego-centric priority structures from local observations and leverages predicted leader behaviors to train a Stackelberg-conditioned actor-critic under CTDE, enabling communication-free execution.

Empirical results across representative dense traffic scenarios demonstrate substantial collision reduction while maintaining competitive system efficiency and smooth control, and show slower performance degradation under increased agent densities. Real-world proof-of-concept trials further demonstrate that the learned local leader-follower relations can

¹www.hiwonder.com/products/jetauto

induce clear yielding and passing behaviors without explicit communication. The successful interaction demonstrates policy feasibility under the nominal sensing noise, actuation disturbances, and control latency of the current testbed. Although TSC is designed for communication-free execution, it can also complement V2X-assisted coordination when communication is available. V2X information may provide richer observations, while the topology-conditioned decision structure can continue to support priority inference and local leader–follower reasoning [39], [40].

Future work will focus on three directions. First, we will systematically evaluate deployment robustness under controlled perception noise, actuation disturbances, sensing and control delays, and occlusion, while further incorporating energy-aware objectives, heterogeneous agent dynamics and intentions, and explicit safety constraints or formal guarantees. Second, we will integrate TSC with alternative CTDE actor-critic backbones, including MAPPO and MADDPG, to assess the generality of the proposed coordination mechanism across different policy-optimization frameworks. Third, beyond the current evaluation on a single simulator and a small-scale physical testbed, we will extend validation to additional simulation environments, larger multi-robot platforms, and practical intersection-management systems, including representative industrial rule-based right-of-way schemes.

REFERENCES

- [1] S. Wang, Q. Li, L. Han, and J. Yan, “A multi-agent reinforcement learning-based longitudinal and lateral control of cavs to improve traffic efficiency in a mandatory lane change scenario,” *Transportation Research Part C: Emerging Technologies*, vol. 161, p. 104558, 2024.
- [2] A. Khalil, K. F. Aljanaideh, and M. Al Janaideh, “On connected autonomous vehicles with unknown human driven vehicles effects using transmissibility operators,” *IEEE Transactions on Automation Science and Engineering*, vol. 20, no. 3, pp. 1876–1889, 2022.
- [3] Y. Liu, B. Xie, Y. Long, J. Chen, and G. Xu, “Learning-based heterogeneous autonomous vehicles scheduling for on-demand last-mile transportation,” *IEEE Transactions on Automation Science and Engineering*, 2025.
- [4] M. Al-Sharman, L. Edes, B. Sun, V. Jayakumar, H. Tahir, M. A. Daoud, B. J. Emran, D. Rayside, and W. Melek, “Autonomous driving at unsignalized intersections: A review of decision-making challenges and reinforcement learning-based solutions,” *IEEE Transactions on Automation Science and Engineering*, vol. 23, pp. 1434–1461, 2025.
- [5] X. Meng and C. G. Cassandras, “Eco-driving of autonomous vehicles for nonstop crossing of signalized intersections,” *IEEE Transactions on Automation Science and Engineering*, vol. 19, no. 1, pp. 320–331, 2020.
- [6] X. Wang and Z. Jin, “Collaborative optimization of multi-equipment scheduling and intersection point allocation for u-shaped automated sea-rail intermodal container terminals,” *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 7373–7393, 2024.
- [7] X. Liang, H. Wang, Y.-H. Liu, Z. Liu, and W. Chen, “Leader-following formation control of nonholonomic mobile robots with velocity observers,” *IEEE/ASME Transactions on Mechatronics*, vol. 25, no. 4, pp. 1747–1755, 2020.
- [8] S. Chen, G. Li, H. Xu, X. Liang, and Y. Zhu, “Marl-based decision making for twin-vehicles cooperative driving at unsignalized intersection,” in *Proceedings of the IEEE International Intelligent Transportation Systems Conference*. IEEE, 2022, pp. 1253–1258.
- [9] J. Xu, P. Hu, and B. Alrifaee, “Sigmarl: A sample-efficient and generalizable marl framework for motion planning,” in *Proceedings of the IEEE International Intelligent Transportation Systems Conference*, 2024.
- [10] A. Ji, R. Wang, S. Wang, and Q. Li, “A marl approach for cooperative ramp merging in mixed traffic,” in *Proceedings of the Engineering Applications of Artificial Intelligence*, 2025.
- [11] A. Aksjonov and V. Kyrki, “Rule-based decision-making system for autonomous vehicles at intersections with mixed traffic environment,” in *Proceedings of the IEEE International Intelligent Transportation Systems Conference*. IEEE, 2021, pp. 1566–1573.
- [12] N. Li, J. Li, and Z. Wang, “Game-theoretic modeling of multi-vehicle interactions at uncontrolled intersections,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 2, pp. 1782–1793, 2022.
- [13] J. Liu, J. Gong, J. Liu, C. Deng, and Y. Lu, “Cooperative decision-making for cavs at unsignalized intersections with right-of-way collaboration awareness,” *IEEE Transactions on Intelligent Transportation Systems*, 2023.
- [14] K. Shu, Z. Deng, R. Liu, and J. Ma, “Decision making in urban traffic: A game theoretic approach for autonomous driving vehicles,” *IEEE Transactions on Intelligent Transportation Systems*, 2025.
- [15] C. Sun, J. Leng, and B. Lu, “Interactive left-turning of autonomous vehicles at uncontrolled intersections,” *IEEE Transactions on Automation Science and Engineering*, vol. 21, no. 1, pp. 204–214, 2022.
- [16] L. Bal, S. Mbakop, G. Espindola-Winck, C. Sueur, and R. Merzouki, “Cooperative curve-based synchronized control of a fleet of autonomous robots,” *IEEE/ASME Transactions on Mechatronics*, vol. 30, no. 4, pp. 2900–2909, 2025.
- [17] R. Lowe, Y. Wu, A. Tamar, J. Harb, P. Abbeel, and I. Mordatch, “Multi-agent actor-critic for mixed cooperative-competitive environments,” in *Proceedings of the Advances in Neural Information Processing Systems*, 2017.
- [18] T. Rashid, M. Samvelyan, C. De Turck, J. Mattar, J. Foerster, and S. Whiteson, “Qmix: Monotonic value function factorisation for deep multi-agent reinforcement learning,” in *Proceedings of the International Conference on Machine Learning*, 2018.
- [19] J. Foerster, Y. Assael, N. de Freitas, and S. Whiteson, “Counterfactual multi-agent policy gradients,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, no. 1, 2018.
- [20] J. Feng, X. Zhao, H. Liu, H. Wang, and Z. Wu, “Right-of-way based marl for collaborative decision-making at unsignalized intersection,” *Expert Systems with Applications*, vol. 262, p. 124671, 2025.
- [21] Z. Zheng and S. Gu, “Safe multi-agent reinforcement learning with bilevel optimization in autonomous driving,” *IEEE Transactions on Artificial Intelligence*, 2024.
- [22] D. Ding, G. Hu, H. Zhou, J. Chen, Y. Li, Z. Li, Z.-Q. Zhang, Y. Xu, and Q. Wei, “Provably efficient generalized lagrangian policy optimization for safety-constrained marl,” in *Proceedings of the Advances in Neural Information Processing Systems*, 2023.
- [23] J. Xu, O. Sobhy, and B. Alrifaee, “Xp-marl: Auxiliary prioritization in marl to address non-stationarity,” *arXiv preprint arXiv:2409.11852*, 2024.
- [24] Y. Liu, C. Wang, H. Xu, and J. Zhang, “A safe marl algorithm using constraint update projection,” *Engineering Applications of Artificial Intelligence*, 2025.
- [25] L. Lin, T. Chen, Y. Wang, and Z. Li, “Robust multi-agent reinforcement learning for autonomous vehicle in noisy highway environments,” in *Proceedings of the Asian Conference on Machine Learning*, 2025.
- [26] B. Zhang, X. Li, J. Pan, and M. Zhang, “Inducing stackelberg equilibrium through spatio-temporal sequential decision-making in marl,” in *Proceedings of the International Joint Conference on Artificial Intelligence*, 2023.
- [27] Zhang, Boyu and Li, Xiaofei and Pan, Jiaqi and Zhang, Min, “Sequential asynchronous action coordination in multi-agent systems: A stackelberg decision transformer approach,” in *Proceedings of the International Conference on Machine Learning*, 2024.
- [28] Y. Zhu, K. Zhang, X. Ma, and S. Li, “A bi-hierarchical game-theoretic approach for network-wide traffic signal control using trip-based data,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 10, pp. 18 783–18 797, 2022.
- [29] H. Hu, M. Tognon, Z. Al-Omaishi, S. Liu, and J. Chen, “Who plays first? optimizing the order of play in stackelberg games with many robots,” in *Proceedings of the Robotics: Science and Systems*, 2024.
- [30] X. Zhang, H. Liu, P. Xu, Z. Chen, and R. Li, “Coupled vehicle-signal control based on stackelberg game enabled multi-agent reinforcement learning,” *Physica A: Statistical Mechanics and Its Applications*, vol. 639, p. 129759, 2025.
- [31] C. Mavrogiannis, J. DeCastro, and S. Srinivasa, “Abstracting road traffic via topological braids: Analysis of interaction complexity and priority structures,” *The International Journal of Robotics Research (WAFR Special Issue)*, 2023.
- [32] C. Mavrogiannis, J. A. DeCastro, and S. S. Srinivasa, “Analyzing multiagent interactions in traffic scenes via topological braids,” in *2022 International Conference on Robotics and Automation (ICRA)*, 2022.

- [33] H. Liu, S. Chen, Y. Wang, J. Chen, F. Ma, and Y. Zhu, “Reasoning multi-agent behavioral topology for interactive autonomous driving,” in *Proceedings of the Advances in Neural Information Processing Systems*, 2024.
- [34] L. Xiao, Z. Pan, Z. Wang, Z. Cao, and W. Li, “Srefiner: Soft-braid attention for multi-agent trajectory refinement,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 960–969.
- [35] T. Mascetta, E. I. Liu, and M. Althoff, “Rule-compliant multi-agent driving corridor generation using reachable sets and combinatorial negotiations,” in *Proceedings of the IEEE Intelligent Vehicles Symposium*, IEEE, 2024.
- [36] Y. Chen, Y. Li, and M. Liu, “From static to dynamic: A survey of topology-aware perception in autonomous driving,” in *Proceedings of the International Conference on Computer Vision Workshops*, 2025.
- [37] K. Dresner and P. Stone, “A multiagent approach to autonomous intersection management,” *Journal of Artificial Intelligence Research*, vol. 31, pp. 591–656, 2008.
- [38] Z. Zhong, M. Nejad, and E. E. Lee, “Autonomous and semi-autonomous intersection management: A survey,” *IEEE Intelligent Transportation Systems Magazine*, 2020.
- [39] Z. Deng, Y. Shi, and W. Shen, “V2X-Lead: LiDAR-based end-to-end autonomous driving with vehicle-to-everything communication integration,” in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023.
- [40] G. Shah, D. Tian, E. Moradi-Pari, and Y. P. Fallah, “AROW: V2X-based automated right-of-way algorithm for cooperative intersection management,” *arXiv preprint arXiv:2304.04958*, 2023.
- [41] F. He, Y. Huang, and H. Wang, “iplan: Interactive and procedural layout planning,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 7793–7802.
- [42] D. Martinez-Baselga, E. Sebastián, E. Montijano, L. Riazuelo, C. Sagüés, and L. Montano, “Avocado: Adaptive optimal collision avoidance driven by opinion,” *IEEE Transactions on Robotics*, 2025.
- [43] M. Bettini, R. Kortvelesy, J. Blumenkamp, and A. Prorok, “Vmas: A vectorized multi-agent simulator for collective robot learning,” *The 16th International Symposium on Distributed Autonomous Robotic Systems*, 2022.
- [44] R. Lowe, Y. Wu, A. Tamar, J. Harb, P. Abbeel, and I. Mordatch, “Multi-agent actor-critic for mixed cooperative-competitive environments,” *Proceedings of the Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [45] T. Rashid, M. Samvelyan, C. S. De Witt, G. Farquhar, J. Foerster, and S. Whiteson, “Monotonic value function factorisation for deep multi-agent reinforcement learning,” *Journal of Machine Learning Research*, vol. 21, no. 178, pp. 1–51, 2020.
- [46] C. Mavrogiannis, J. DeCastro, and S. S. Srinivasa, “Analyzing multi-agent interactions in traffic scenes via topological braids,” in *2022 International Conference on Robotics and Automation (ICRA)*. IEEE, 2022, pp. 5806–5813.
- [47] J. S. Birman and T. E. Brendle, “Braids: a survey,” in *Handbook of knot theory*. Elsevier, 2005, pp. 19–103.
- [48] M. Klimke, B. Völz, and M. Buchholz, “Automatic intersection management in mixed traffic using reinforcement learning and graph neural networks,” *arXiv preprint arXiv:2301.12717*, 2023.
- [49] C. Chen, M. Cai, J. Wang, K. Li, Q. Xu, J. Wang, and K. Li, “Cooperation method of connected and automated vehicles at unsignalized intersections: Lane changing and arrival scheduling,” *arXiv preprint arXiv:2109.14175*, 2021.
- [50] Y. Yang, R. Luo, M. Li, M. Zhou, W. Zhang, and J. Wang, “Mean field multi-agent reinforcement learning,” in *Proceedings of the International Conference on Machine Learning*. PMLR, 2018, pp. 5571–5580.
- [51] S. Nayak, K. Choi, W. Ding, S. Dolan, K. Gopalakrishnan, and H. Balakrishnan, “Scalable multi-agent reinforcement learning through intelligent information aggregation,” in *Proceedings of the International Conference on Machine Learning*. PMLR, 2023, pp. 25 817–25 833.

APPENDIX A

THEORETICAL ANALYSIS OF

STACKELBERG-CONDITIONED VALUE LEARNING

A.1 Bellman Error Bound Under Leader-Prediction Noise

Fix a follower agent i and its local leader set $\mathcal{L}_i^t$ defined in Sec. III-B3. Let T_i^{SE} denote the Bellman evaluation operator of

the local Stackelberg MDP in (38), where leader actions are the realized actions $a_j^t \sim \pi_\theta(\cdot | u_j^t)$ for all $j \in \mathcal{L}_i^t$. Let $T_i^{\text{SE,pred}}$ denote an auxiliary operator obtained by replacing these leader actions in the Bellman backup by the corresponding predictions $\hat{a}_{j|i}^t$ for all $j \in \mathcal{L}_i^t$. Let V_i^{SE} and $V_i^{\text{SE,pred}}$ be the unique fixed points of T_i^{SE} and $T_i^{\text{SE,pred}}$, respectively.

Lemma 1 (Bellman error bound). Assume the per-step reward is bounded in magnitude by $|r_i^t| \leq R_{\max}$, so any discounted value function under these operators satisfies $\|V\|_\infty \leq V_{\max}$ with $V_{\max} = R_{\max}/(1 - \gamma)$. Suppose that for some $\varepsilon_T \geq 0$ we have

$$\|T_i^{\text{SE}}V - T_i^{\text{SE,pred}}V\|_\infty \leq \varepsilon_T \quad (48)$$

for all bounded V with $\|V\|_\infty \leq V_{\max}$. Then

$$\|V_i^{\text{SE}} - V_i^{\text{SE,pred}}\|_\infty \leq \frac{\varepsilon_T}{1 - \gamma}. \quad (49)$$

Proof. Recall that both T_i^{SE} and $T_i^{\text{SE,pred}}$ are γ -contractions in the sup norm. Using $V_i^{\text{SE}} = T_i^{\text{SE}}V_i^{\text{SE}}$ and $V_i^{\text{SE,pred}} = T_i^{\text{SE,pred}}V_i^{\text{SE,pred}}$, we obtain

$$\begin{aligned} \|V_i^{\text{SE}} - V_i^{\text{SE,pred}}\|_\infty &= \|T_i^{\text{SE}}V_i^{\text{SE}} - T_i^{\text{SE,pred}}V_i^{\text{SE,pred}}\|_\infty \\ &\leq \|T_i^{\text{SE}}V_i^{\text{SE}} - T_i^{\text{SE,pred}}V_i^{\text{SE}}\|_\infty \\ &\quad + \|T_i^{\text{SE,pred}}V_i^{\text{SE}} - T_i^{\text{SE,pred}}V_i^{\text{SE,pred}}\|_\infty \\ &\leq \varepsilon_T + \gamma\|V_i^{\text{SE}} - V_i^{\text{SE,pred}}\|_\infty, \end{aligned} \quad (50)$$

where the first term is bounded by assumption and the second term uses the contraction property of $T_i^{\text{SE,pred}}$. Rearranging gives $(1 - \gamma)\|V_i^{\text{SE}} - V_i^{\text{SE,pred}}\|_\infty \leq \varepsilon_T$, which yields (49). $\square$

Lemma 1 shows that if the Bellman backup error caused by leader-action prediction is uniformly small, then the induced value $V_i^{\text{SE,pred}}$ is uniformly close to the ideal local Stackelberg value V_i^{SE} , with error amplified by at most the factor $1/(1 - \gamma)$.

A.2 Policy Improvement in the Local Stackelberg MDP

For a follower policy π_i and fixed leader policies $\pi_{\mathcal{L}_i^t} = \{\pi_j\}_{j \in \mathcal{L}_i^t}$, the local Stackelberg follower objective is

$$\begin{aligned} J_i^{\text{SE}}(\pi_i; \pi_{\mathcal{L}_i^t}) &= \mathbb{E} \left[\sum_{k=0}^{\infty} \gamma^k r_i^{t+k} \mid a_i^{t+k} \sim \pi_i(\cdot | u_i^{t+k}), \right. \\ &\quad \left. a_j^{t+k} \sim \pi_j(\cdot | u_j^{t+k}), \forall j \in \mathcal{L}_i^t \right], \end{aligned} \quad (51)$$

where the expectation is taken over trajectories induced by $(\pi_i, \pi_{\mathcal{L}_i^t})$. Let $V_{i,\pi}^{\text{SE}}$ and $Q_{i,\pi}^{\text{SE}}$ denote the corresponding value and action value functions, and define the advantage

$$A_{i,\pi}^{\text{SE}}(u, a) = Q_{i,\pi}^{\text{SE}}(u, a) - V_{i,\pi}^{\text{SE}}(u). \quad (52)$$

Lemma 2 (Performance difference lemma for the local Stackelberg MDP). Consider the local Stackelberg MDP of agent i with fixed leader policies $\pi_{\mathcal{L}_i^t} = \{\pi_j\}_{j \in \mathcal{L}_i^t}$ at a given time step t and its associated leader set $\mathcal{L}_i^t$. For any two follower policies $\pi_i, \tilde{\pi}_i$ and the corresponding discounted state visitation distribution d_{π_i} induced by $(\pi_i, \pi_{\mathcal{L}_i^t})$, we have

$$\begin{aligned} J_i^{\text{SE}}(\pi_i; \pi_{\mathcal{L}_i^t}) - J_i^{\text{SE}}(\tilde{\pi}_i; \pi_{\mathcal{L}_i^t}) &= \\ &= \frac{1}{1 - \gamma} \mathbb{E}_{u \sim d_{\pi_i}, a \sim \pi_i(\cdot | u)} [A_{i,\pi_i}^{\text{SE}}(u, a)], \end{aligned} \quad (53)$$

where

$$d_{\pi_i}(u) = (1 - \gamma) \sum_{k=0}^{\infty} \gamma^k \Pr(u_i^{t+k} = u \mid \pi_i, \pi_{\mathcal{L}_i^t}). \quad (54)$$

If π_i is deterministic, $a = \pi_i(u)$, we may write $A_{i,\tilde{\pi}_i}^{\text{SE}}(u) := A_{i,\tilde{\pi}_i}^{\text{SE}}(u, \pi_i(u))$ and obtain

$$J_i^{\text{SE}}(\pi_i; \pi_{\mathcal{L}_i^t}) - J_i^{\text{SE}}(\tilde{\pi}_i; \pi_{\mathcal{L}_i^t}) = \frac{1}{1 - \gamma} \mathbb{E}_{u \sim d_{\pi_i}} [A_{i,\tilde{\pi}_i}^{\text{SE}}(u)]. \quad (55)$$

Proof. Fix the leader policies $\pi_{\mathcal{L}_i^t}$. Let $V_{i,\tilde{\pi}_i}^{\text{SE}}$ be the value function under $(\tilde{\pi}_i, \pi_{\mathcal{L}_i^t})$. For any state u and action a , define

$$Q_{i,\tilde{\pi}_i}^{\text{SE}}(u, a) = \mathbb{E} \left[r_i^t + \gamma V_{i,\tilde{\pi}_i}^{\text{SE}}(u_i^{t+1}) \mid u_i^t = u, a_i^t = a, \pi_{\mathcal{L}_i^t} \right]. \quad (56)$$

By the definition of advantage,

$$A_{i,\tilde{\pi}_i}^{\text{SE}}(u, a) = \mathbb{E} \left[r_i^t + \gamma V_{i,\tilde{\pi}_i}^{\text{SE}}(u_i^{t+1}) - V_{i,\tilde{\pi}_i}^{\text{SE}}(u_i^t) \mid u_i^t = u, a_i^t = a, \pi_{\mathcal{L}_i^t} \right]. \quad (57)$$

Now consider trajectories generated by $(\pi_i, \pi_{\mathcal{L}_i^t})$ starting from time t . Taking conditional expectation of the previous identity at each time $t + k$ and summing with weights γ^k gives

$$\begin{aligned} \mathbb{E} \left[\sum_{k=0}^{\infty} \gamma^k A_{i,\tilde{\pi}_i}^{\text{SE}}(u_i^{t+k}, a_i^{t+k}) \right] &= \mathbb{E} \left[\sum_{k=0}^{\infty} \gamma^k r_i^{t+k} + \right. \\ &\quad \left. \sum_{k=0}^{\infty} \gamma^{k+1} V_{i,\tilde{\pi}_i}^{\text{SE}}(u_i^{t+k+1}) - \sum_{k=0}^{\infty} \gamma^k V_{i,\tilde{\pi}_i}^{\text{SE}}(u_i^{t+k}) \right]. \end{aligned} \quad (58)$$

The two value sums telescope, leaving

$$\mathbb{E} \left[\sum_{k=0}^{\infty} \gamma^k A_{i,\tilde{\pi}_i}^{\text{SE}}(u_i^{t+k}, a_i^{t+k}) \right] = J_i^{\text{SE}}(\pi_i; \pi_{\mathcal{L}_i^t}) - J_i^{\text{SE}}(\tilde{\pi}_i; \pi_{\mathcal{L}_i^t}), \quad (59)$$

where the second term appears because under $(\tilde{\pi}_i, \pi_{\mathcal{L}_i^t})$, $V_{i,\tilde{\pi}_i}^{\text{SE}}(u_i^t)$ equals the discounted return starting from state u_i^t .

Finally, rewrite the left-hand side using the discounted visitation distribution. By definition of d_{π_i} ,

$$\begin{aligned} \mathbb{E} \left[\sum_{k=0}^{\infty} \gamma^k A_{i,\tilde{\pi}_i}^{\text{SE}}(u_i^{t+k}, a_i^{t+k}) \right] &= \\ &= \frac{1}{1 - \gamma} \mathbb{E}_{u \sim d_{\pi_i}, a \sim \pi_i(\cdot|u)} [A_{i,\tilde{\pi}_i}^{\text{SE}}(u, a)], \end{aligned} \quad (60)$$

which yields (53). The deterministic case (55) follows by substituting $a = \pi_i(u)$. $\square$

Corollary 1. Let $V_{\phi}^i(u_i^t, \{\hat{a}_{j|i}^t\}_{j \in \mathcal{L}_i^t})$ denote the Stackelberg-conditioned critic in (35). If V_{ϕ}^i matches the predicted local Stackelberg state-value $V_i^{\text{SE,pred}}(u_i^t)$ on the support of the discounted visitation distribution d_{π_i} , then the policy-gradient update based on advantages estimated from V_{ϕ}^i is a stochastic gradient-ascent step on the predicted follower objective $J_i^{\text{SE,pred}}(\pi_i; \pi_{\mathcal{L}_i^t})$ induced by $T_i^{\text{SE,pred}}$. Moreover, when the induced bias is controlled as in Lemma 1, this update provides a tractable approximation to improving the ideal

follower objective $J_i^{\text{SE}}(\pi_i; \pi_{\mathcal{L}_i^t})$. In particular, for fixed higher-priority neighbors $\mathcal{L}_i^t$, the actor update with the Stackelberg-conditioned critic realizes a local Stackelberg follower best response, while the evolving leader policies $\pi_{\mathcal{L}_i^t}$ implicitly specify the upper level of a decentralized bilevel Stackelberg optimization.

APPENDIX B IMPLEMENTATION DETAILS

B.1 Environment, Observation, and Reward Specification

We instantiate the simulation environment based on VMAS [43] and SigmaRL [9]. This appendix specifies the agent dynamics, local observation, action space, and reward function used in our experiments.

B.1.1 Agent Dynamics: Each agent is modeled as a non-linear kinematic bicycle model with state

$$\xi_i^t = (p_{x,i}^t, p_{y,i}^t, \theta_i^t), \quad (61)$$

where $(p_{x,i}^t, p_{y,i}^t)$ is the planar position and θ_i^t is the heading angle. The action is

$$a_i^t = (v_i^t, \delta_i^t), \quad v_i^t \in [-v_{\max}, v_{\max}], \quad \delta_i^t \in [-\delta_{\max}, \delta_{\max}], \quad (62)$$

where v_i^t is the commanded speed and δ_i^t is the steering angle. Let L_f and L_r be the front and rear half-wheelbase. The slip angle is

$$\beta_i^t = \arctan \left(\frac{L_r}{L_f + L_r} \tan \delta_i^t \right), \quad (63)$$

and the continuous-time dynamics are

$$\begin{aligned} \dot{p}_{x,i} &= v_i^t \cos(\theta_i^t + \beta_i^t), \\ \dot{p}_{y,i} &= v_i^t \sin(\theta_i^t + \beta_i^t), \\ \dot{\theta}_i &= \frac{v_i^t}{L_f + L_r} \cos(\beta_i^t) \tan \delta_i^t. \end{aligned} \quad (64)$$

The dynamics are integrated by RK4 with sampling interval Δt . We use the same vehicle model and dynamics parameters as [9].

B.1.2 Observation Design: At time t , agent i receives an ego-centric local observation

$$o_i^t = \{\tilde{o}_i^t, \{o_{i,j}^t\}_{j \in \mathcal{N}_i^t}\}, \quad (65)$$

where $\tilde{o}_i^t$ is the self-observation and $o_{i,j}^t$ is the observation of neighbor j . The self-observation is

$$\tilde{o}_i^t = \left(v_i^{t,\parallel}, d_{i,\text{CL}}^t, d_{i,\text{LB}}^t, d_{i,\text{RB}}^t, \{q_{i,k}^t\}_{k=1}^{n_{\text{p,RP}}} \right) \in \mathbb{R}^{4+2n_{\text{p,RP}}}, \quad (66)$$

where $v_i^{t,\parallel}$ is the forward velocity in the ego frame; $d_{i,\text{CL}}^t$, $d_{i,\text{LB}}^t$, and $d_{i,\text{RB}}^t$ are the distances from the current position of agent i to the lane center line, left boundary, and right boundary, respectively; and $q_{i,k}^t \in \mathbb{R}^2$ is the k -th point on the short-term reference path. Here, $n_{\text{p,RP}}$ denotes the number of sampled short-term reference-path points. In our experiments, $n_{\text{p,RP}} = 3$, yielding $\tilde{o}_i^t \in \mathbb{R}^{10}$.

All geometric quantities in the observation are expressed in the ego coordinate frame of agent i , whose origin is located at the current position of agent i and whose x -axis is aligned with its heading. Thus, the global position and heading of agent i

are implicit, and neighbor information is encoded as relative geometry and motion with respect to agent i .

For each neighbor $j \in \mathcal{N}_i^t$, we use

$$o_{i,j}^t = (\{q_{i,j,k}^t\}_{k=1}^4, \Delta v_{i,j}^t, d_{i,j}^t) \in \mathbb{R}^{11}, \quad (67)$$

where $q_{i,j,k}^t \in \mathbb{R}^2$ is the k -th vertex of neighbor j 's rectangular footprint in agent i 's ego frame, $\Delta v_{i,j}^t \in \mathbb{R}^2$ is the relative velocity, and $d_{i,j}^t$ is the inter-agent distance. Specifically, $\Delta v_{i,j}^t$ is represented in agent i 's ego frame, and $d_{i,j}^t$ denotes the Euclidean distance between agents i and j .

B.1.3 Action Design: The policy outputs a diagonal Gaussian distribution over $a_i^t = (v_i^t, \delta_i^t)$. The sampled action is squashed by tanh and rescaled to $[-v_{\max}, v_{\max}] \times [-\delta_{\max}, \delta_{\max}]$.

B.1.4 Reward Design: The per-agent reward consists of path-following, efficiency, safety, and smoothness terms. Each coefficient is normalized by $Z = 100$, i.e., $c(\cdot) = c_{(\cdot)}^{\text{raw}}/Z$, and negative coefficients indicate penalties.

Progress. Let $\Delta \tilde{p}_i^t$ be the displacement of agent i from $t-1$ to t , represented in the ego frame of agent i , and let ω_k be normalized decreasing weights for the short-term reference points. The progress reward is

$$r_{i,\text{prog}}^t = \frac{c_{\text{prog}}}{v_{\max} \Delta t} \sum_{k=1}^{n_{\text{p,RP}}} \omega_k \langle \Delta \tilde{p}_i^t, q_{i,k}^t \rangle, \quad (68)$$

which rewards displacement aligned with the ego-frame short-term reference path.

Speed. The speed reward is

$$r_{i,\text{vel}}^t = \eta_i^t c_{\text{vel}} \frac{v_i^{t,\parallel}}{v_{\max}}, \quad \eta_i^t = \begin{cases} 1, & v_i^{t,\parallel} > 0, \\ 2, & v_i^{t,\parallel} \leq 0, \end{cases} \quad (69)$$

which encourages forward motion along the reference direction and penalizes reverse motion more strongly.

Agent proximity. Let $f_{\text{exp}}(d; \tau_{\text{lo}}, \tau_{\text{hi}})$ be an exponentially decreasing proximity function over the distance range $[\tau_{\text{lo}}, \tau_{\text{hi}}]$. The proximity penalty is

$$r_i^{t,\text{prox}} = c_{\text{na}} \sum_{j \neq i} f_{\text{exp}}(d_{i,j}^t; \tau_{\text{agt,lo}}, \tau_{\text{agt,hi}}), \quad (70)$$

where $d_{i,j}^t$ is the distance between agents i and j .

Path deviation. With $d_{i,\text{CL}}^t$ denoting the distance to the lane center line and τ_{dev} the deviation threshold,

$$r_i^{t,\text{dev}} = c_{\text{dev}} \frac{d_{i,\text{CL}}^t}{\tau_{\text{dev}}}, \quad (71)$$

which penalizes deviation from the assigned reference path.

Steering smoothness. Let $\tilde{\delta}_i^t \in [-1, 1]$ be the normalized steering command and τ_{δ} the steering-change threshold. The steering penalty is

$$r_i^{t,\text{steer}} = c_{\text{steer}} \frac{\max(0, |\tilde{\delta}_i^t - \tilde{\delta}_i^{t-1}| \delta_{\max} - \tau_{\delta})}{2\delta_{\max} - 2\tau_{\delta}}, \quad (72)$$

which is activated only for abrupt steering changes.

Collisions.

$$r_i^{t,\text{col}} = c_{\text{ca}} \mathbb{I}[\text{collide}_{i,\text{agent}}^t] + c_{\text{cb}} \mathbb{I}[\text{collide}_{i,\text{map}}^t], \quad (73)$$

where the two indicators denote collisions with other agents and map boundaries.

Direction-aware time term.

$$r_i^{t,\text{time}} = c_{\text{time}} \text{sign}(v_i^{t,\parallel}) \frac{\|v_i^t\|}{v_{\max}}. \quad (74)$$

This term rewards nonzero speed in the reference direction and penalizes backward motion, discouraging both reverse driving and overly conservative stopping.

Total reward. The final reward is clipped to stabilize training:

$$r_i^t = \text{clip}\left(r_i^{t,\text{prog}} + r_i^{t,\text{vel}} + r_i^{t,\text{prox}} + r_i^{t,\text{dev}} + r_i^{t,\text{steer}} + r_i^{t,\text{col}} + r_i^{t,\text{time}}, -1, 1\right). \quad (75)$$

The reward coefficients are kept consistent with SigmarL [9] to ensure a fair comparison.

B.2 Network Implementation Details

Table V summarizes the module-level implementation of TSC-Net. The network follows an ego-centric design. Ego and neighbor observations are first encoded separately and then fused into a topology-aware latent representation for priority inference, action prediction, and Stackelberg-conditioned value estimation. All agents share the same network parameters under CTDE.

APPENDIX C

ADDITIONAL EXPERIMENTAL RESULTS

C.1 Comparative Training Curves

To further assess convergence behavior and robustness to random initialization, we compare the training curves of TSC, XP-MARL, SigmaRL, and MFPO in the *Clover* training scenario over five independent random seeds. Fig. 10 reports the agent-agent collision rate and episode mean reward as functions of environmental steps. Each curve represents the mean across the five seeds, and the shaded region denotes one standard deviation.

As shown in Fig. 10, TSC reduces the agent-agent collision rate more rapidly and converges to a lower level than the compared methods. It also exhibits a faster increase in episode mean reward and stabilizes at a higher value. The bounded variation across the five random seeds further indicates that the learned policy is robust to random initialization and maintains reliable convergence.

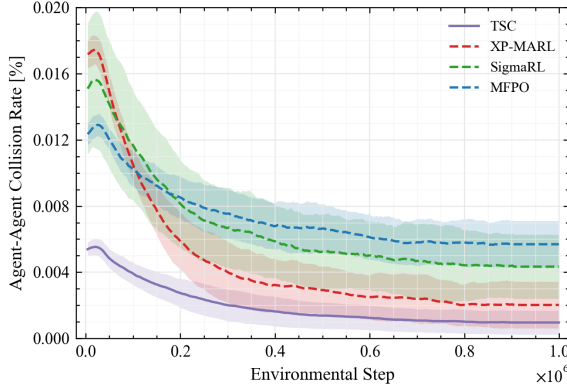
C.2 Training Dynamics of Ablation Variants

Fig. 11 compares the training dynamics of TSC and its ablation variants in the *Clover* scenario. Each curve shows the mean over five independent random seeds, with the shaded region indicating one standard deviation. After the initial exploration stage, the complete TSC model achieves the lowest agent-agent collision rate and the highest converged episode reward, indicating the most favorable overall learning behavior among the evaluated variants.

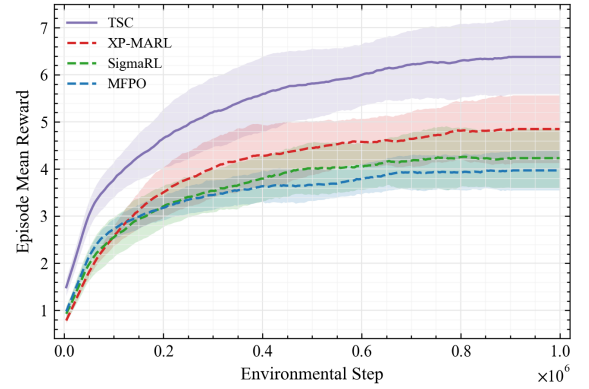
Removing the Top-K filter yields the highest residual collision rate and the lowest reward, suggesting that retaining

TABLE V: Network architecture of TSC-Net.

Module	Input	Output	Architecture	Usage
EgoEncoder	$\tilde{o}_i^t \in \mathbb{R}^{10}$	$h_i^t \in \mathbb{R}^{128}$	MLP with hidden layers (256, 256) and Tanh activations	Encodes ego features.
NeighborEncoder	$o_{i,j}^t \in \mathbb{R}^{11}$	$h_{i,j}^t \in \mathbb{R}^{128}$	MLP with hidden layers (256, 256) and Tanh activations	Encodes neighbor features.
TopoEncoder	$\mathbf{H}_i^t \in \mathbb{R}^{ \mathcal{N}_i^t \times (d_e + d_n)}$	$\mathbf{Q}_i^{\text{topo}} \in \mathbb{R}^{ \mathcal{N}_i^t \times 128}$	Neighbor-shared MLP with hidden layers (512, 256) and ReLU activations	Encodes pairwise topology relations.
TopoHead	$\mathbf{Q}_i^{\text{topo}} \in \mathbb{R}^{ \mathcal{N}_i^t \times 128}$	$\hat{\mathbf{p}}_i^t \in [0, 1]^{ \mathcal{N}_i^t }$	Neighbor-shared MLP with hidden layer (64) and element-wise sigmoid output	Predicts edge-wise priorities.
NodeHead	$\mathbf{Q}_i^{\text{topo}} \in \mathbb{R}^{ \mathcal{N}_i^t \times 128}$	$\hat{s}_i^t \in \mathbb{R}$	Masked pooling followed by MLP with hidden layer (64), ReLU activation, and linear output	Predicts node priority.
TopoAttention	$h_i^t, \mathbf{H}_{i,K}^t, \hat{\mathbf{p}}_{i,K}^t$	$C_i^t \in \mathbb{R}^{128}$	Single-head cross-attention over Top- K neighbors	Aggregates priority-filtered neighbors.
EgoDecoder	$[C_i^t, \hat{s}_i^t] \in \mathbb{R}^{129}$	$u_i^t \in \mathbb{R}^{128}$	MLP with hidden layers (256, 256) and Tanh activations	Forms decision state.
PolicyHead	$u_i^t \in \mathbb{R}^{128}$	$\mu_i^t, \sigma_i^t \in \mathbb{R}^2$	MLP with hidden layers (256, 256) and Tanh activations, followed by Gaussian parameter extraction	Outputs action distribution.
PredictHead	$\mathbf{H}_{i,K}^t, \hat{\mathbf{p}}_{i,K}^t$	$\{\hat{a}_{j i}^t\}_{j \in \mathcal{N}_{i,K}^t}, \hat{a}_{j i}^t \in \mathbb{R}^2$	Neighbor-shared MLP with hidden layer (64), ReLU activation, and linear output	Predicts neighbor actions.
ValueHead	$u_i^t, \{\hat{a}_{j i}^t\}_{j \in \mathcal{L}_i^t}$	$V_\phi^i \in \mathbb{R}$	MLP with hidden layers (256, 256) and Tanh activations	Estimates state value.



(a) Agent-agent collision rate.



(b) Episode mean reward.

Fig. 10: Comparative training curves of TSC and the learning-based baselines in the *Clover* training scenario. The curves represent the mean over five independent random seeds, and the shaded regions denote one standard deviation.

all observed neighbors introduces weakly relevant interactions that impede learning. The random-priority variant also performs substantially worse than TSC, indicating that sparsification alone is insufficient without an informative priority structure. Although the Euclidean Top- K variant improves over these two variants, it still converges to a lower reward and a higher collision rate than TSC, showing that geometric proximity does not always identify the neighbors most relevant to crossing and merging conflicts.

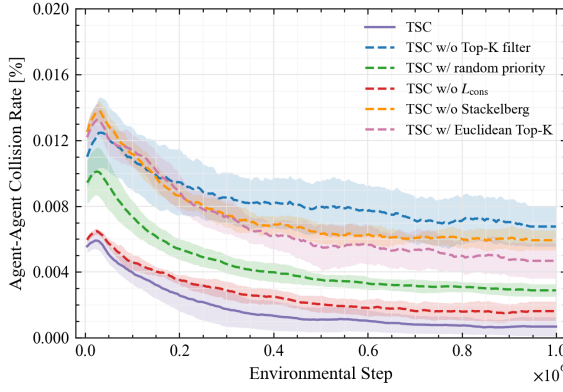
Removing Stackelberg-conditioned value reasoning degrades both collision reduction and reward convergence, supporting the role of local leader-follower conditioning in coordinated policy learning. The variant without $\mathcal{L}_{\text{cons}}$ performs closest to the complete model, but still converges to a lower reward and a higher collision rate, confirming the benefit of consistency between edge-wise and node-wise priority predictions. Together with the post-training results in Table III, these curves show that the proposed components improve both final performance and learning convergence.

C.3 Leader-Action Prediction Accuracy

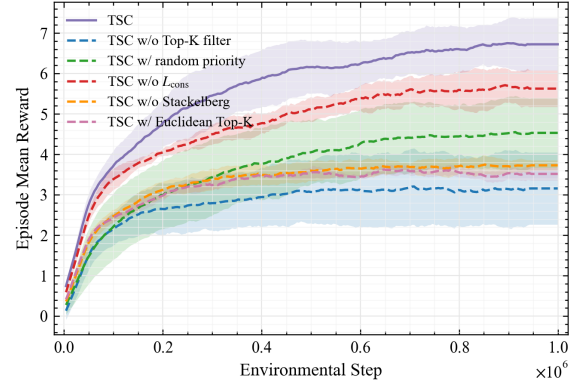
To quantitatively assess the reliability of the leader-action predictions used by the Stackelberg-conditioned critic, we evaluate the one-step prediction error during testing on held-out trajectories. For each random seed, the RMSE is computed over all predicted actions of the inferred local leaders, and the final results are reported as the mean and standard deviation across five independent seeds. Both speed and steering commands are normalized to $[-1, 1]$.

TABLE VI: One-step leader-action prediction RMSE on held-out trajectories over five random seeds. The actions are normalized to $[-1, 1]$.

Scenario	Speed RMSE ↓	Steering-Angle RMSE ↓
Clover	0.0093 ± 0.0075	0.1051 ± 0.0546
Weave	0.0175 ± 0.0118	0.0527 ± 0.0249
Merge	0.0040 ± 0.0016	0.0956 ± 0.0566
Bypass	0.0238 ± 0.0099	0.0622 ± 0.0325



(a) Agent-agent collision rate.



(b) Episode mean reward.

Fig. 11: Training dynamics of TSC and its ablation variants in the *Clover* scenario. (a) Agent-agent collision rate. (b) Episode mean reward. Each curve represents the mean over five independent random seeds, and the shaded region denotes one standard deviation.

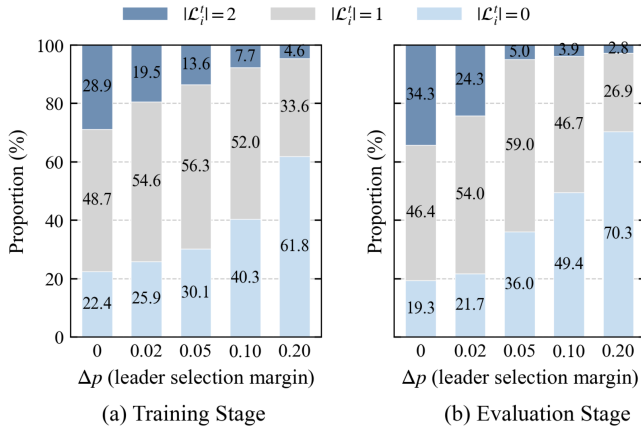


Fig. 12: Distribution of local leader-set sizes under different leader-selection margins Δp . The proportions are averaged over all valid agent-time pairs and five random seeds. Training statistics are computed from the final 20% of training steps, and evaluation statistics are obtained using the final policies.

TABLE VII: Single-agent decentralized policy inference latency of TSC-Net with a batch size of one.

Metric	Latency (ms)
Mean $\pm$ standard deviation	0.63 ± 0.11
95th percentile	0.80

As shown in Table VI, the prediction module achieves consistently low speed RMSE across all four scenarios, ranging from 0.0040 to 0.0238. The steering RMSE is moderately higher, ranging from 0.0527 to 0.1051, due to the greater variability of steering commands in interactive situations. Nevertheless, the prediction errors remain limited relative to the normalized action range, supporting the use of predicted leader actions for critic-side value estimation.

C.4 Distribution of Local Leader-Set Sizes

To further assess the effect of the leader-selection margin Δp , we analyze the distribution of the local leader-set size $|L_i^t|$. Since the Top- K neighborhood size is fixed at two, $|L_i^t| \in \{0, 1, 2\}$.

As shown in Fig. 12, increasing Δp reduces the frequency of multi-leader cases while increasing the proportion of empty leader sets. At $\Delta p = 0$, samples with $|L_i^t| = 2$ account for 28.9% during training and 34.3% during evaluation. In contrast, $\Delta p = 0.20$ raises the proportion of empty leader sets to 61.8% and 70.3%, respectively, indicating that an overly large margin can suppress leader conditioning.

The default setting $\Delta p = 0.05$ provides a balanced and sparse local hierarchy. Single-leader cases account for 56.3% of the training samples and 59.0% of the evaluation samples, while multi-leader cases remain limited to 13.6% and 5.0%, respectively. The similar distributions observed during training and evaluation further indicate that the inferred leader structure remains consistent under the final policy.

C.5 Policy Inference Latency

To characterize the execution-time efficiency of TSC-Net, we measure the single-agent decentralized policy inference latency with a batch size of one on the same hardware described in Sec. IV-B. Input states are uniformly sampled from the four evaluation scenarios. The measured forward pass includes local feature encoding, priority inference, Top- K selection, topology-conditioned aggregation, and policy decoding, while the critic-side modules used only during centralized training are excluded.

As reported in Table VII, the mean inference latency is 0.63 ± 0.11 ms, with a 95th-percentile latency of 0.80 ms. Both are well below the 50 ms control interval used in our experiments, indicating that the decentralized policy computation can be completed within each control cycle. This measurement characterizes policy-network inference rather than the complete sensing-to-actuation latency.